General Biology (Botany and Genetics)

Biology Module

Course Name: Biology 121

Course code: BIO 121

Lilongwe University of Agriculture and Natural Resources

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Unit 1: Introduction to Botany

Introduction

As you may be aware by now that the field of biology is divided into several branches based on the subject of study. In Biology module 1, you covered microbiology (study of microorganisms) and zoology (study of animals). In this module you will be introduced to other two branches of biology which are botany (the study of plants) and genetics (the study of heredity).

Unit 1 learning objectives

By the end of this Unit, you should be able to:

a)

Unit 1 key terms

Bryophyta

Pteridophyta

Angiosperms

Gymnosperms

Tracheids

Gametangia

Sporogenous

Autotrophs

Sporophyte

Gametophyte

Alternation of generations

Archeoplastids

Strobili

Thalloid

Resin

Needle

Unit 1 content

1.0 Plant systematics

There are about 1.8 million species of organisms and several millions more remain to be discovered. To study these species, a system for organizing, naming and classifying them is essential. The field of biology that deals with the diversity of living organisms and their evolutionary relationships is called **systematics**. **Taxonomy** is the branch of science of dealing with the naming, describing and classifying of organisms. On the other hand, **classification** refers to the branch of science involving grouping of organisms based on their evolutionary similarities and differences. Carolus Linnaeus devised the binomial nomenclature of naming organisms which uses the genus and the species. The genus name always starts with a capital letter followed by the species name which always starts with a small letter. In printed form the scientific name is presented in italics and when it is hand written the genus and the species names must always be underlined separately.

There are eight levels of classification which are hierarchical (from highest to lowest). These levels are Domain, Kingdom, Phylum, Class, Order, Family, Genus, and Species.

Domain refers to the broadest taxonomic category or group to which a living thing belongs; each domain includes one or more kingdoms. As you may recall from module 1, the three domain system consists of Archea (prokaryotic), Eubacteria (prokaryotic) and Eukaryote (eukaryotic). There are five kingdoms to which all organisms belong under these three domains. These kingdoms are Kingdom

Monera, Kingdom **Protista**, Kingdom **Fungi**, Kingdom **Plantae** and Kingdom **Animalia**.

1.1 Kingdom Plantae

The Kingdom plantae is also called as **Kingdom Metaphyta**. It includes all types of eukaryotic, multicellular, photosynthetic and autotrophic plants with differentiated tissues and organs. Most of the organisms in this kingdom are **autotrophs**, which synthesize their own food with the help of solar energy through photosynthesis. However, there are a few species, which are both autotrophs and heterotrophs.

The history of life on earth and the success of many organisms literally depend on the success of plants. The Kingdom Plantae contains about 300,000 different species. Among the five kingdoms, Kingdom plantae is very important, as they are the source of food for all other living creatures present on planet earth, which depends on plants to survive.

Plants evolved from aquatic green algal ancestors known as **charophytes** or stoneworts. Land plants are collectively classified as **archaeplastids**. Adaptations to life on land that have evolved in plants include a waxy cuticle to prevent water loss or desiccation; multicellular gametangia: (organs that protect developing reproductive parts); stomata (tiny openings in the leaves and stems for gaseous exchange); and for most plants, vascular tissues mostly the xylem and phloem for transportation of nutrients and water throughout the plant. Plants undergo an alternation of generations between multicellular gametophyte and sporophyte generations.

1.3 Characteristics and adaptations of plants

Besides the diversity of plants, plants share several characteristics. We will now list the main characteristics that are shared among most plants. Most plants: -

a. are multicellular eukaryotic organisms with differentiated tissues and organs.

b. have cell walls made of cellulose.

c. have cells which frequently contain large vacuole and photosynthetic pigments in plastids.

d. have photosynthetic pigments are chlorophylls *a* and *b*.

e. are nonmotile.

f. reproduce both asexually and sexually, with alternation of gametophyte (*n*) and sporophyte (2*n*) generations.

1.4 Adaptation of plants

Here we will discuss the general adaptations of plants for life on land. Life as we know today started in water and evolved to terrestrial environments. Terrestrial plants had to attain the following adaptations.

- Plants have a waxy cuticle that covers the aerial portion and prevents desiccation or drying out due to evaporation
- They have tiny openings called stomata on the leaves and stems which facilitate gaseous exchange
- They have multicellular sex organs (gametangia) with an outer layer of sterile cells that surrounds and protects the delicate gametes (eggs and sperm cells). The fertilized egg develops into an embryo within the gametangium which

offers protection during development. This is not the case in algae where eggs develop away from the ganetongium and in some algae, the gametes are released before fertilization.

1.5 The plant life cycles and alternation of generations

Plants have two distinctive life cycles namely gametophyte generation and sporophyte generation. These two phases alternate where by the plants spend the first part of their life cycle in a multicellular haploid stage and second part of their life cycle in a multicellular diploid stage.

Gametophyte generation: This is the haploid portion which gives rise to haploid gametes by mitosis.

Sporophyte generation: This is the diploid portion of the life cycle which begins when two gametes formed during the gametophyte generation fuse. The sporophyte generation produces haploid spores by meiosis. These spores represent the first stage in gametophyte generation.

This change during the life cycle from gametophyte generation to sporophyte generation is called **alternation of generations**.

During alternation of generations, the haploid gametophytes produce male gametangia known as Antheridia (singular antheridium) in which sperm cells form and female gametangia known as archegonia (singular archegonium). Each archegonium contains a single egg which is fertilized by a single sperm upon being reached and forms a zygote.

The formation of the diploid zygote marks the beginning of a sporophyte generation. The formed zygote undergoes mitotic cellular division and develops into a multicellular embryo which forms a young sporophyte plant which grows into a mature sporophyte plant. The mature sporophyte has sporogenous cells which are spore producing cells or spore mother cells which divide by meiosis to produce haploid spores.

This can be summarized as follows:

Fertilisation of female sex cell by male sex cell

Zygote

Embryo

Mature sporophyte plant	Mature gametophyte plant
Sporogenous cells	Arhegonis (female sex cell)/
Meiosis	Antheridia (male sex cell)
Spores	

Gametophyte produces gametes (eggs/sperms) by mitosis, two haploid gametes fuse to form a zygote, the zygote develops by mitosis into the sporophyte, special cells of the sporophyte undergo meiosis to form spores and each spore has the potential to undergo mitosis and develop into a gametophyte

The figure below shows the two distinct generations of the basic plant.

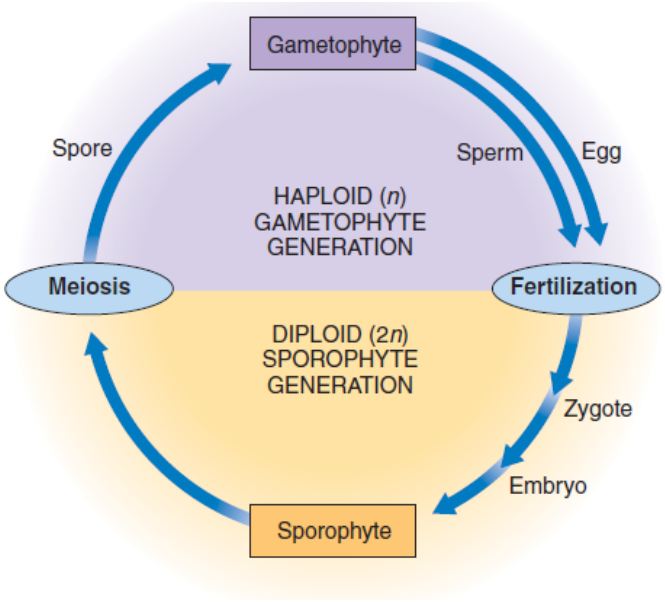
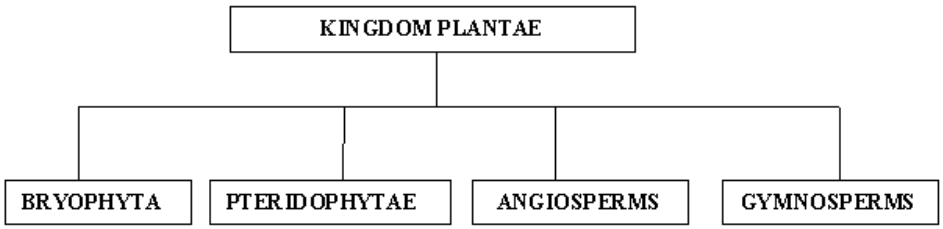


Figure 1.1: The basic plant life cycle

1.6 Classification of plants

Plants are divided into four main groups based on reproduction and vascular tissues. The vascular tissues and reproduction cycle are mainly considered in plant classification due to the significant roles that they play. For Instance vascular tissues (the xylem and the phloem) play a vital role in transporting substances (water, minerals and sugars) throughout the plant while seeds contain an embryo that helps in storing food which is used during germination. The four major groups of plants are Bryophytes, Seedless vascular plants, Gymnosprems and Agniosperms which are seeded vascular plants.

The plant kingdom is made up of 10 phyla namely: Phylum Bryophyta, Phylum Hepatophyta, Phylum Anthocerophyta, Phylum Pteridophyta, Phylum Lycopodiophyta, Phylum Coniferophyta, Phylum cycadophyta, Phylum Ginkgophyta, Phylum Gnetophyta, and Phylum Anthrophyta or Angiosperms. However, the focus of this module is on the four main phyla: **Bryophyta, Pteridophyta, Angiosperms and Gymnosperms.**



1.6.1 Bryophyta

Bryophyta comes from the Greek words *bryon* meaning moss and *phyton* meaning plant. It includes the lichens, mosses, algae and the fungi. Bryophytes are the simplest and the most primitive of the land plants. They are believed to be the earliest plant to colonize terrestrial habitat from aquatic environment They are small nonvascular plants that lack a specialized vascular, or conducting system to transport nutrients, water, and essential minerals (inorganic nutrients) throughout the plant body. Their adaptation to a terrestrial mode of life is partial as water is indispensable in one stage or another in their life cycle. Hence, they are also known as the amphibians of the plant kingdom. They do not form true roots, stems, or leaves. Instead of using the vascular tissues for transportation, these plants depend mainly on osmosis and diffusion. Due to this utilization of osmosis and diffusion for nutrient transportation, bryophytes do not grow as other higher plants. Bryophytes have a restricted growth since some of their cells cannot obtain sufficient necessary materials if they are to grow much larger. These plants occur more commonly during rainy seasons and also in humid areas.

The Bryophytes are nonvascular plants with a dominant gametophyte generation. They are further divided into three distinct phyla which differ in many ways and in terms of evolutionary relatedness they may or may not be closely related. They are grouped together because they lack vascular tissues and they have similar life cycles. Bryophtes are grouped into three phyla names; Phylum Bryophyta

(mosses), Phylum Hepatophyta (liverworts) and Phylum Anthocerophyta (hornworts

The life cycle of bryophytes consists of two distinct phases. These phases are the gametophytic phase and sporophytic phases which alternate with each other a process called alternation of generations. For reproduction they reproduce and disperse through haploid spores. Their life cycle consists of a marked alternation of generations with dominant gametophyte generation.

1. **Gametophytic phase**: It is the main phase which is haploid, autotrophic, gamete formative (male and female) and is responsible for sexual reproduction. The gametophytes generally form dense green mats consisting of individual plants

2. **Sporophytic phase**: It is diploid, heterotrophic, spore formative and responsible for the asexual reproduction. Vegetative reproduction takes place through fragmented tubers adventitious branches and gemmae. Sexual reproduction takes place by sex organs antheridium (male) and archegonium (female).

   Fertilization takes place in the presence pf water. After fertilization, a zygote is formed which undergoes divisions to form an embryo. The figure below shows the life cycle of mosses as an example of bryophytes

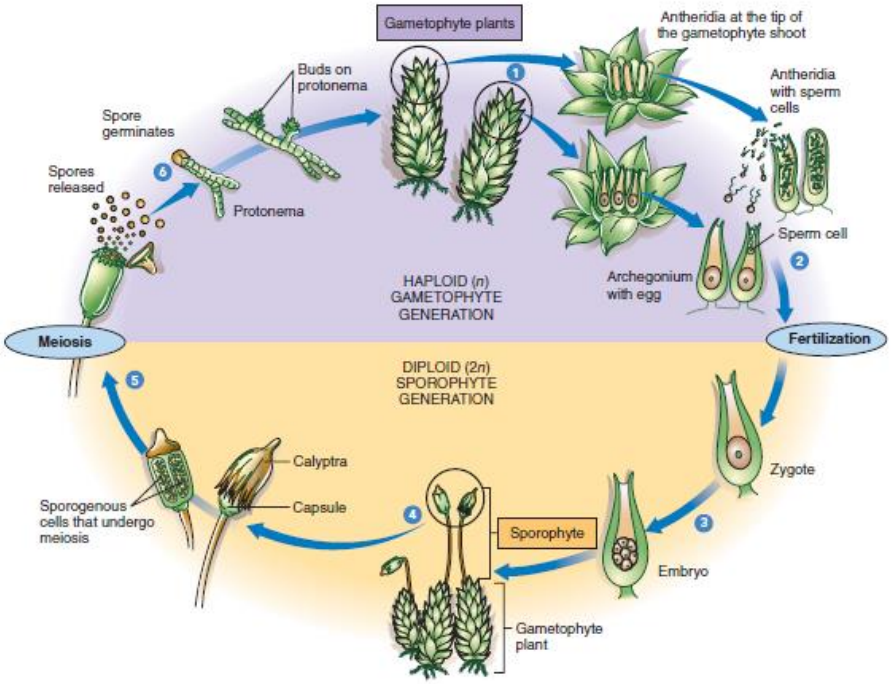


Figure 1.2: The life cycle of moses.

1.6.2 Pteridophyta

These are seedless vascular plants. Evolutionarily, they are the first terrestrial plants to possess vascular tissues – xylem and phloem. The pteridophytes are normally found in swamps, marshes, stream banks, and tropical rain forests. Some may grow in fields, crevices on cliffs or mountains.

The life cycle of pteridophytes has clearaly defined alternation of generations just as in bryophytes. You may recall that in bryophytes the dominant phase in the life cycle is the gametophytic plant body. However, in pteridophytes, the main plant body is a sporophyte which is differentiated into true root, stem and leaves. These organs possess well-differentiated vascular tissues. The leaves in pteridophyta are small (microphylls) as in ferns. The sporophyte has a horizontal underground stem (**rhizome**) that bears leaves known as **fronds.** Spore production occurs in the fronds where sporangia developes. Most species bear the spores in clusters called

sori (singular sorus). In the sporangia, sporagenous cells / spore mother cells under-go meiosis to form haploid spores. The sporangia burst open and release spores that may germinate and develop into gametophytes. A mature fern gametophyte is a tiny heart shaped structure that grows against the ground called a **prothallus** (plural prothalli). The prothallus lacks vascular tissues, it only has tiny hair like absorptive rhizoids for anchorage. The prothallus produces both archegonia and antheridia on its underside. A single egg is contained in each archegonia while numerous sperm cells are produced in each antheridium.

The sporophytes bear sporangia that are held by leaf-like appendages called sporophylls. In some cases sporophylls may form distinct compact structures called **strobili** or **cones**.

The sporangia produce spores by meiosis in spore mother cells. The spores germinate to give rise to very small but multicellular free-living, mostly photosynthetic **thalloid** gametophytes called **prothallus (free living haploid gametophyte)**. These gametophytes require cool, damp, shady places to grow. Because of this specific restricted requirement and the need for water for fertilization, the spread of pteridophytes is limited and restricted to narrow geographical regions. The gametophytes bear male and female sex organs called **antheridia** and **archegonia**, respectively just as in bryophytes.

Water is required for transfer of **antherozoids** – the male gametes released from the antheridia, to the mouth of **archegonium**. Fusion of a male gamete with the egg present in the archegonium result in the formation of a diploid zygote that grows into a multicellular embryo. This multicellular embryo is an immature sporophyte which is attached to the gametophyte and depends on the gametophyte. As the embryo matures, the prothallus withers and dies and the prothallus becomes free living. Thus the pteridophyte life cycle alternates between

the dominant sporophyte and which jas rhizome, roots, and fronds and the haploid gametophyte (prothallus).The sporophyte generation is dominant over the gametophyte generation not only because it is large than the gametophyte but because it also persists over extended periods. A good example of the pteridophytes are the ferns whose sporophytes are perennials while the gametopytes die soon after reproducing.

In summary the pteridophytes

- Are seedless vascular plants, which contain vascular tissues but do not produce seeds.
- Have their reproduction process carried out by spores
- Have a life cycle that alternates between the gametophyte and sporophyte whereby the sporophyte dominates
- Some pteridophytes e.g. ferns and club mosses are **heterosporous** : produce two types of spores which are microspores and mega spores unlike bryophytes which are monosporous- producing only one type of spores

The figure below shows the life cycle of n example of a pteridophyte, the fern.

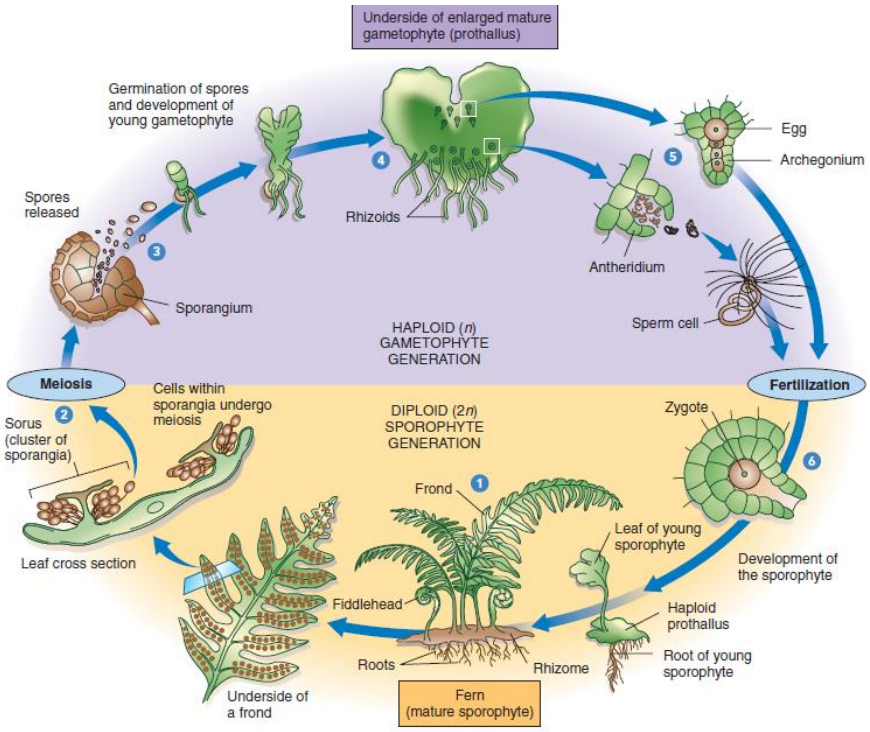


Figure 1.3: A typical life cycle of a pteridophyte

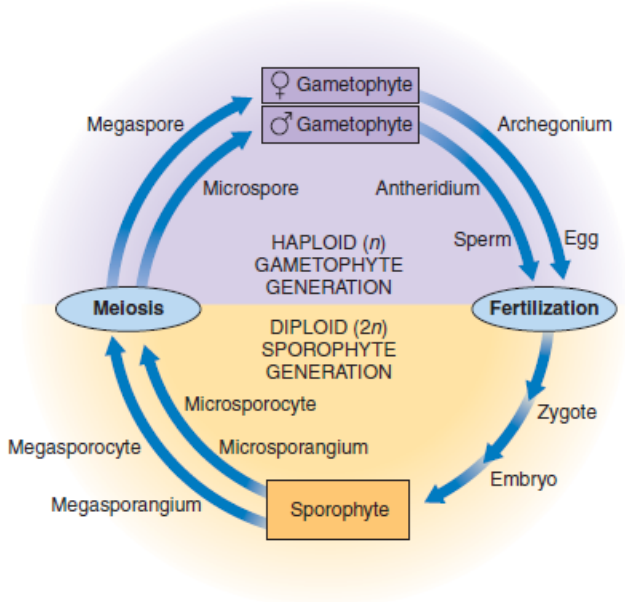


Figure 1.4: The basic life cycle of heterosporous plants

Summary

1. Sporophyte plant produces sporangia in conelike strobilus containing microsporangia and megasporangia. Microsprongia produce microsporocytes / mother spore cells which undergo meiosis to form microscopic haploid microspores.
2. Microspores develop into male gametophyte that produces sperm cells within antheridia
3. Megasporangia undergoes meiosis to form haploid megaspores that develop into female gametophyte to produce eggs in archegonia
4. Fertilization occurs followed by development of new sporophyte.

1.6.3 Gymnosperms

The gymnosperms are classified imto four phyla:coniferophyta, commonoy called conifers; Ginkgophyta,Cycadophyta and Gnetophyta. The conifers are the most familiar group of gymnosperms and they consist of pines, spruces, hemlocks and firs. The conifers are woody plants that produce seed in cones.

Characteristics of conifers

- Conifers' wood has **tracheids** which are long tapering cells with pits through which water and dissolved minerals move from one cell to the other.
- They produce a viscous clear or translucent substance called **resin**; consisting of several organic compounds which protects the plant from insect and fungi attack.
- They have long, narrow, tough, and leathery leaves called **needles**.
- Most conifers are evergreen, apart from a very few that are deciduous shed their needles at the end f he growing season.

- Most conifers are **monoecious**, they have separate male and female reproductive parts in different locations on the same plant.

Gymnosperms are non-flowering plants with undeveloped seeds, which are present in an enclosed structure. In other terms they are plants in which the ovules are not enclosed by any ovary wall and remain exposed before and after fertilization. They produce naked seeds. The gymnosperms are heterosporous; producing haploid microspores and megaspores. The two kinds of spores are produced within sporangia that are borne on sporophylls which are arranged spirally along an axis to form lax or compact **strobili** or **cones**. The strobili bearing microsporophylls and microsporangia are called **microsporangiate** or **male strobili.** The microspores develop into a male gametophytic generation which is highly reduced and is confined to only a limited number of cells. This reduced gametophyte is called a **pollen grain**. The development of pollen grains takes place within the microsporangia. The cones bearing megasporophylls with ovules or megasporangia are called **macrosporangiate** or **female strobili**. The male or female cones or strobili may be borne on the same plant (trees). However in some cases male cones and megasporophylls are borne on different plants (trees).

Unlike bryophytes and pteridophytes, in gymnosperms the male and the female gametophytes do not have an independent free-living existence. They remain within the sporangia retained on the sporophytes. The pollen grain is released from the microsporangium. They are carried in air currents and come in contact with the opening of the ovules borne on megasporophylls. The pollen tube that carries the male gametes grows towards archegonia in the ovules and discharge their contents near the mouth of the archegonia. Following fertilization, a diploid zygote develops into an embryo and the ovules into seeds. These seeds are not covered.

1 Monocot begins with a single seed-leaf. The main veins of their leaves are usually parallel and unbranched.

2 Monocot plays an important role in providing us with our primary sources of nutrition, which includes grains, fruits, etc.

1.6.4 Angiosperms

These are flowering plants, which develop the seeds within a protective structure. They are the most diverse phylum of plants that form flowers for sexual reproduction and they produce seeds enclosed in fruits. There are two main types of angiosperms: monocots and eudicots/ dicot

Main characteristics of Angiosperms

- They develop their seeds within an ovary, which itself is embedded in a flower. After the stage of fertilization, the flower falls and the ovary bulges to become a fruit.
- Angiosperms in the class Dicotyledoneae grows into two seed-leaves (cotyledons).
- An angiosperms leaf consists of a single, branched, main vein, which originates from the base of the leaf blade. In few plats, it may also consist of four or more main veins diverging from the same base.

Unit 1 activity

Unit 1 summary

Unit 1 test

Unit 2: Morphology and Anatomy of Flowering Plants

Introduction

Naturally, all flowering also known as seed plants develop from a seed. The seed is the seat of partial development of the new **sporophyte**, the embryo and it thus plays a major role in providing the continuity between successive generations of the plants. Given the right environmental factors which include moisture, temperature, nutrients and light the seed will germinate. This becomes possible after a more or less prolonged **dormancy** caused by various internal factors such as rudimentary or physiologically immature embryos, mechanically resistant or impermeable seed coats, or the presence of germination inhibitors.

Germination is essentially a resumption of embryo growth after imbibition of water. During imbibition, the water content of the seed rises, usually fast at first, then slower, and the embryonic tissue becomes metabolically active. **Enzymes** already present are activated and new proteins with specific enzymatic activities are synthesized for the digestion and utilization of the different kinds of stored materials. Cell extension and cell divisions are initiated and proceed according to the programmed pattern. This growth requires continuous supply of water and nutrients. Before the embryo becomes a self-supporting seedling it utilizes the food stored in the **endosperm** and/or the embryo itself.

Seedling growth leads to the development of three major **vegetative organs,** i.e. roots, stem and leaf. Depending on the type of plants, time taken to grow into a full mature plant

will vary. At maturity, these plants produce flowers (**reproductive organs**) which lead to the development of fruits and seeds.

In this unit, we are going to study different **morphological** and **anatomical** structures of these organs of the plants. You will also be introduced to the important roles, variations and similarities of these organs.

Unit 2 learning outcomes

By the end of this unit, you should be able to:

1. State the primary functions and types of roots, stems, leaves and flowers.
2. Describe the external and internal parts of mature primary root, stems, leaves and flowers.
3. Compare morphological and anatomical structures of monocotyledonous and dicotyledonous stems and leaves.

Unit 2 key terms

Sporophyte
Dormancy
Imbibition
Enzymes
Endosperm
Androecium
Herbaceous
Differentiation
Coalescence
Inflorescence
Gymnoecium
Vegetative organs
Reproductive organs
Morphology
Anatomy

Unit 2 Content

2.1 Roots

2.1.1 Functions of roots

In this section we will discuss the four major and primary functions of roots. You may have noticed that some plants might use roots for other specialized functions. Among major functions of the roots include anchorage, absorption, transportation, and translocation. Specialized functions include storage, propagation, respiration, balancing and support, growth, parasitism, and association with other living organisms. We will begin our discussion by looking at the main function of the roots.

Anchorage

This should be familiar to most of us that roots anchor trees firmly into the soil, usually through an extensive branching network that constitutes about one third of the total dry weight of the plant. The roots of most plants do not extend more than 3 to 5 meters down into the earth; those of many **herbaceous** species are confined to the upper 0.6 to 0.9 meters. Some plants form very shallow root systems, but these systems still effectively anchor the plants with a densely branching mass of roots radiating out in all directions as far as 15 meters from the stem.

Absorption

You may also be familiar that roots absorb water and minerals in solution mostly through feeder roots found in the soil.

Transportation

After absorbing the water and mineral salts from the soil roots are also responsible for transporting these materials through stems to the leaves by the use of vascular bundles laid in a form of continuous pipes forming a body known as **xylem**. This is a one way direction process from roots to leaves (Figure 1A).

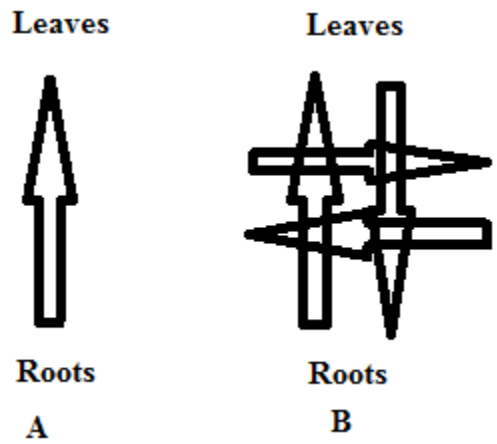


Figure 2.1: Transportation of water, one way direction through xylem pipes (A), and translocation of manufactured food, multidirectional process through phloem pipes (B).

Translocation

Food manufactured in the leaves is supposed to be transported to plant parts where it is required. As in the transportation of water, there are also specialized pipes that are laid through the plant body stretching from roots to leaves. Such pipes are known as **phloem**. Manufactured food is able to be moved from leaves to roots and back from roots to upper parts of the plant where it is required. Manufactured food can also be moved sideways through the root and stem bodies to wherever it is required, hence the process is multidirectional (Figure 2B).

2.1.2 Types of root systems

Based on the origin, position, function and morphological appearance of the root, roots can be grouped into three major groups of taproot, adventitious or fibrous and modified or specialized root systems. Let us now describe these root systems.

Tap root system

This is characterized by one large primary root (taproot) and smaller roots with varying sizes called secondary roots which originate from the primary root. The taproot originates from the embryonic root called **radicle** (Figure 2) and continues to form the root systems of many dicotyledonous plants for example bean, mango, orange, gmalina, and gymnosperms for example pines.

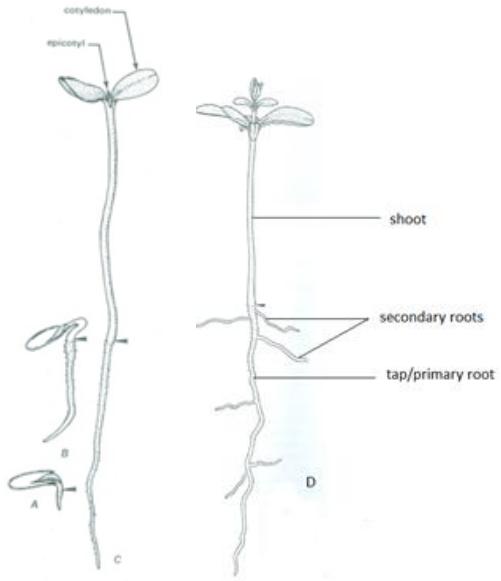


Figure 2.2: Development of taproot system from embryonic radicle (A, B, C, D)

Adventitious or fibrous root system

On the other hand you will notice that roots of monocotyledonous plants such as maize, rice, wheat, grasses and bamboos and dicotyledonous plants that are planted from stems for example cassava, sugarcane and sweet potatoes form adventitious systems.

In the monocotyledons the tap root commonly lives for a relatively short time and the root system is formed by roots arising directly from the shoot (stem and leaves) often in connection with axillary buds. Adventitious root system usually has many uniform or homogenous roots and in appearance, they tend to look like a broom or a mop. Such a fibrous broom structures is the origin of its name as fibrous root system (Figure 3).

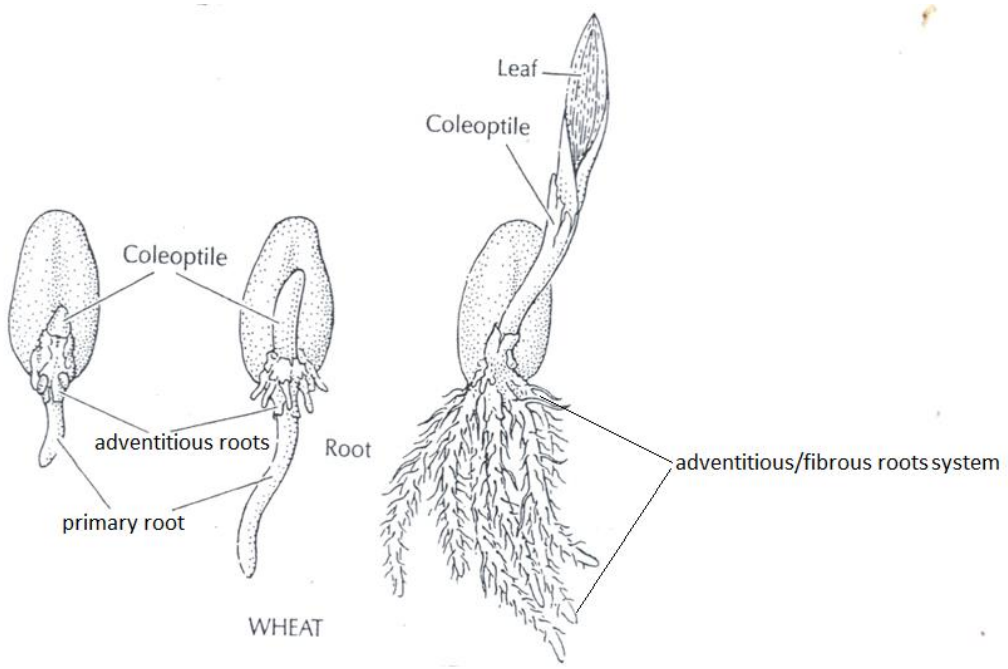


Figure 2.3: Adventitious or fibrous root system

Modified or specialized root systems

As mentioned earlier, most plants produce either adventitious or fibrous root system, taproot system or more commonly combinations of the two types. Some plants, however have roots with modifications that adapt them for performing specific functions apart from the four basic functions. Some of the examples of the modified root systems are given in Table 1 below.

Table 1: Examples of modified roots

Modified/specialized root	Examples
1. Freshy roots (Food and water storage roots)	Cassava, sweet potatoes, yams, and some members of pumpkin and plantain familes.
2. Propagative roots – many plants produce adventitious buds (buds not appearing from the stem but other organs of the plant ie roots growing along the surface and leaves which will lead to production of a new plant.	Adventitious roots of ricepaper plants (*Tetrapanax papyrifera*), Horseradish (*Roroppa armoracia*), roots of cherries, apples, pears and other fruit trees often produce suckers.
3. Aerial roots a. Includes respiratory or pneumatophores used for gas exchange in plants growing in swampy environments. b. prop roots are roots developing from stems to provide extra support	Black mangrove (*Avicennia nitida*), yellow water weed (*Ludwigia repens*) -maize and other grasses, fig trees (*Ficus sp*)
4. Association with other living organisms a. mycorrhizae: plant roots associated with fungi for mutual symbiotic purposes. b. nodules in leguminous roots: mutualistic symbiotic association between roots and bacteria c. parasitic association: an association where roots of crops or trees are associated with roots of plant parasitic plants.	-roots of most forest trees and grasses -all roots of leguminous plants -witchweed (*Striga sp*), Dodders, orchids.

2.1.3 Root structure

You will appreciate that the root has different internal and external structures that makes the root to perform the different functions. As the roots elongates, it displays varying morphological and anatomical structural features. From the growing tip to the mature part of the root, we observe six different structural regions (Figure 4).

Such regions include root cap, apical meristem zone, elongation zone, differentiation zone, root hair zone and mature root zone. These are discussed below in relation to their type of cells and functions.

Root tip

The root tip is the growing point of the root. As the name suggests this is found at the tip of the root. Within the root tip you will find the root cap, the apical meristem, the elongation zone and the region of cell differentiation.

Root cap

The root cap is composed of parenchyma cells that produce and exude mucus-like substance. This substance is smeared outside the root cap and acts as lubricating oil as the root pushes through the soil particles. As you will notice, the root cap protects the apical meristem. The root cap also assists the growing root as it penetrates the soil.

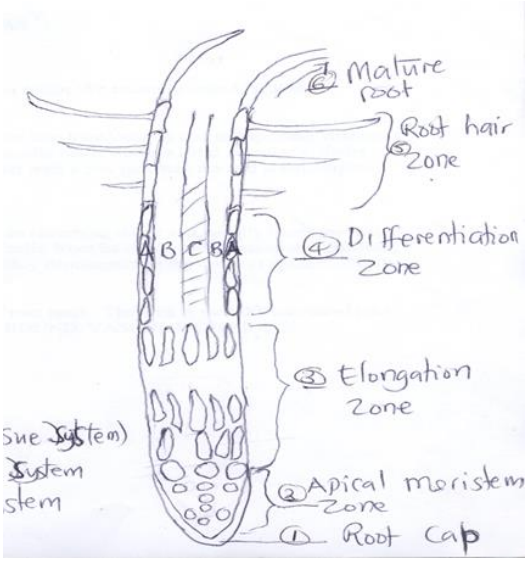


Figure 2.4: Internal and external structure of a root

Apical meristem zone

The apical meristem is a zone is made up of young and undifferentiated cells (meristems). These cells are randomly arranged. The meristems are capable of dividing mitotically increasing the numbers in a limited area hence pushing the root tip forward (growing).

Elongation zone

In the elongation zone, cells start elongating and increase in size. As the cells elongate they arrange themselves in chains

Differentiation zone

In this zone cells begin to differentiate into different tissue. The cells begin by forming primary tissues for both structural and functional use. The first primary tissues include dermal tissue (epidermis), ground tissue (cortex) and vascular tissue (xylem, phloem and pericycle) (Figure 4). The outermost cells arrange themselves into a skin like structure called **dermal tissue** (epidermis) while the innermost cells organise themselves into translocating and transporting structures called **vascular tissues**. The vascular tissue forms a solid vascular cylinder, or **pith** is present then a hollow cylinder with a few parenchyma and sclerenchyma cells embedded in them for support. The cells between the dermal and the vascular tissue form ground tissues made of three major zones that is hypodermis or exodermis, loose cortex and endodermis.

Root hair zone

In young roots the epidermis is specialized as an absorbing tissue and usually bears root hairs which are tubular extensions of the epidermal cells. Root hairs markedly extend the absorbing surface of the root. Root hairs have short life; they disintegrate in the older epidermal cells as they reach mature stage.

Mature root zone

In this region you will find fully grown up cells that make the mature body of the root are well separated and differentiated in readiness for their different functions they have to undertake in the root. They are seen to be grouped into three main groups (Figure 5). The peripheral is made up of dermal tissue which acts as a protective skin for the root. The central part of the body is taken up by a cylindrical shaped vascular tissues that are meant to transport water and mineral salts and translocate manufactured food. The area between the two is formed by cells grouped as ground tissues. Their functions are many, among them include storage, support and in aiding transportation and translocation.

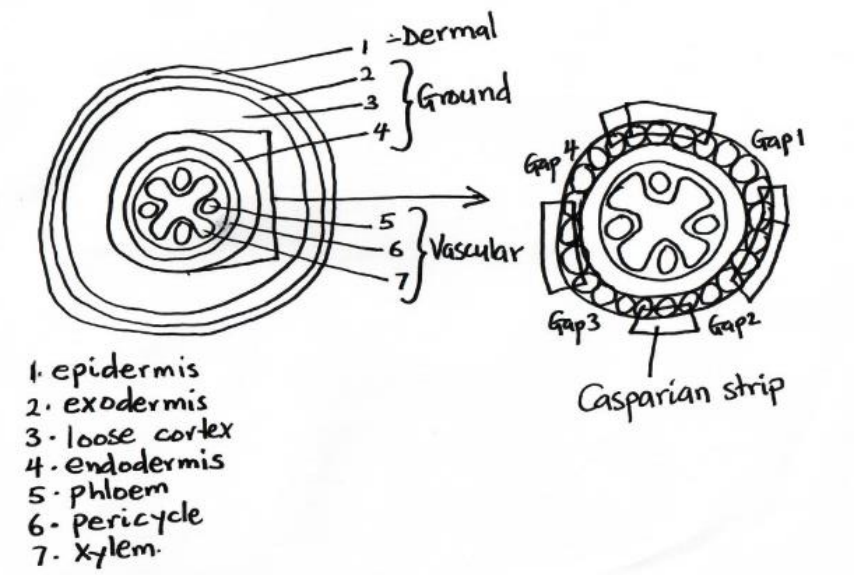


Figure 2.5: Cross section of a mature root

Dermal tissue

Epidermis is the outermost one cell layer body which acts as a protective skin of the root. At this stage, you will notice that most cells have lost their root hairs and seen to be forming a very tightly held layer to prevent entry of water and other large molecules from the soil. The layer is composed of epidermal cells of varying sizes.

Ground tissues

Ground tissue is a group of tissues that occupy the middle section of the body of a mature root. The ground tissue is mostly composed of parenchyma cells but they could also have collenchyma cells. The tissue is characterised with large intercellular spaces (lacunae) formed by **aerenchyma** cells. Aerenchyma cells are highly vacuolated and may contain plastids are devoid of chlorophyll. They however as act storage for large amounts of starch.

The innermost layer of the ground tissue is differentiated into one cell layer called **endodermis**. The outermost layer close to the dermal tissue (epidermis) is made up of one or more cell layers called **exodermis**. The large layer between exodermis and endodermis is composed of loosely packed cells which usually are referred to as **loose cortex**.

Cells that form the endodermis have their cell walls coated with a waterproof substance called **suberin.** This substance goes round the cells within the radial and transversal wall forming a band-like region. However, the band does not completely cover all cells. A few cells are left free allowing direct openings between cortex and the vascular tissues. It is through these openings that water and soil solutions find their way into the vascular tissues. The band is called **casparian strip**. This band is important in the transportation of water and soil solutions between cortex and vascular tissues, where water enters the vascular tissue pipes through pressure and with the same pressure water and soil solutions are taken up to the leaves.

Vascular tissue

Let us now finish our discussion of the root by looking at the vascular tissue. The vascular tissue form a cylinder-like body of the innermost part of the mature root (vascular cylinder). This cylinder has three main parts namely pericycle, xylem and phloem. We will discuss the composition, positions and functions of each of these parts below.

Pericycle

The pericycle is a one celled layer. The cells are either Parenchymatous or sclerenchymatous. The pericycle acts as a source for lateral (secondary) roots, vascular cambium and cork cambium (phellogen).

Xylem

The xylem forms a solid core with ridge like projections called **archs.** The number of archs varies in different species. Among roots of the same plants and in relation to this variation the roots are called **Diarch** with two projections, **Triarch** with three projections, **Tetrarch** with four projections or **Polyarch** if the root has_many projections.

In general roots of dicotyledonous plants have 2, 3, or 4 archs (poles) and in monocotyledonous plants they may have many archs.

Xylem pipes with varying sizes form a continuous pipe system from roots through stems to the leaves. These pipes are responsible for the transportation of water and mineral salts from roots to the leaves.

Phloem

Phloem pipes are grouped into bundles between the xylem archs. Like in the xylem, these pipes also form a continuous system stretching from roots to leaves. Their major function is to translocate or conduct manufactured food from leaves throughout the plant body.

2.2 Stems

2.2.1 Functions of stems

We will begin by learning the main functions of plant stems. Stems have five main functions which are a) carry, support and expose leaves to light and air b) transportation of water and mineral salts c) translocation of manufactured food d) storage of nutrients and e) perform metabolic functions such as photosynthesis and respiration.

2.2.2 External structure

Hope you will agree that the close association of the stem with the leaves make the aerial part of plant axis strictly more complex than the root. Shoot means stems and leaves as one system which seems to express the association. In contrast to the root, the shoot has nodes and internodes with one or more leaves per node. Depending on the development of the internodes the shoot assumes different aspects.

Visible or invisible internodes

Plants with well-developed internodes tend to have visible stems as in beans, sunflower, maize, bamboos, gmelina, fruit trees and forest trees, while some plants like onions and other bulbs and bananas have short invisible or hidden stems as their internodes are not well developed.

One or more auxiliary buds per node

Another variation is the arrangement of leaves per node on the stem which will lead to number of auxiliary buds hence number of the lateral shoots. Some plants have one leaf per node hence one axillary bud; others have two leaves per node leading to two axillary buds and yet others have three or more leaves per node leading to three or more axillary buds.

The differences are also associated with type of growth and habitat of the shoot. Air or underground stems for rhizome like in grasses, bulbs like in onions, corms like in banana and tubers as in potatoes. Typical examples of land or water shoots are rice plants and the water hyacinth. Some shoots are upright while others are climbing or creeping. Vines creeping shoots, stolons are also creeping shoots as in grasses while some varieties of beans are climbers (Figure 6).

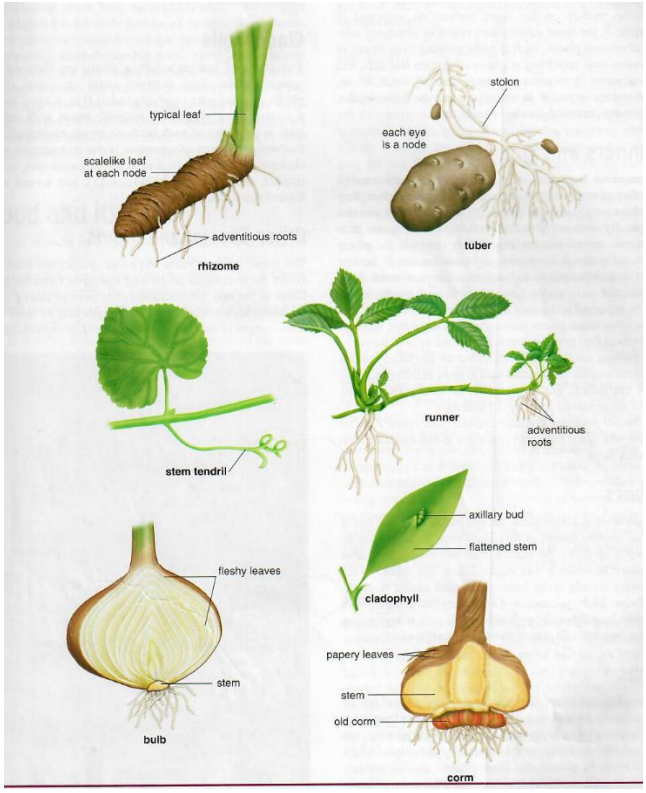
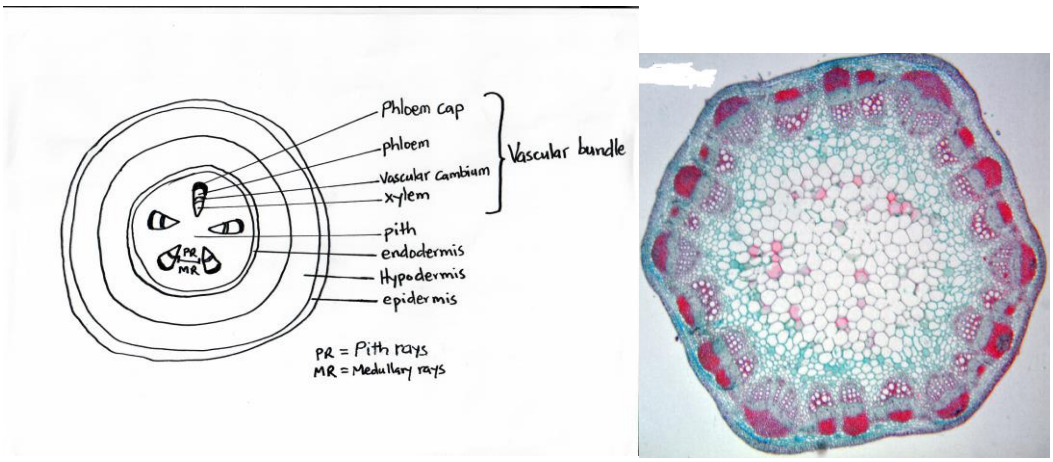


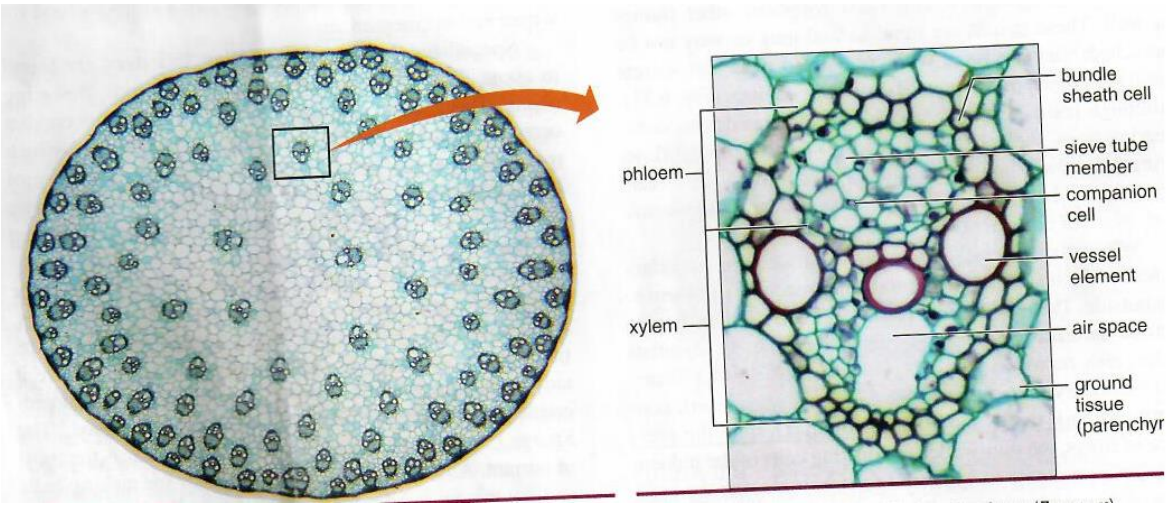
Figure 2.6: Modified stems

2.2.3 Internal structure of primary stem

The stem like the root consists of three tissue systems namely dermal, ground and vascular tissues. However the distribution of ground and vascular tissues vary between stems of dicotyledonous and monocotyledonous plants. In stems of dicotyledonous and gymnospermae plants, vascular tissues appear in separate bundles and these bundles are arranged in a circular structure within the ground tissue. The inner part of the ground tissue is called pith and the outer one cortex. In stems of monocotyledonous plants, vascular tissues form vascular bundles which are randomly distributed within the ground tissue where the pith and cortex cannot easily be separated (Figure 7 a and b).



a



b

Figure 2.7: a) Cross section of dicotyledonous b) cross section of a monocotyledonous stem

Dermal tissue

The dermal tissue in stems continues to be a one celled layer as we have already seen in roots and it is called epidermis. This skin is maintained up to maturity in all cotyledonous plants and herbaceous dicotyledonous plants. Examples of the herbaceous dicotyledonous plants include bean, peas, sunflower and other small leguminous plants. However, in trees and other larger dicotyledonous plants, another thicker skin called **periderm** replaces epidermis at maturity.

Ground tissue

Unlike in the root, the ground tissue has four major parts. The innermost part is called pith and close to the dermal is called cortex with three different layers: hypodermis, loose cortex and endodermis. In stems the endodermal layer does not perform the function as in the root and its cells do not contain suberin. They are mainly used to store starch hence referred to as starch cells.

Vascular tissue

Primary vascular system of stems of seed plants consist of strands variable in size and degree of distinctness. Individual strands are referred to as **vascular bundles**. Each bundle is composed of xylem and phloem pipes. Xylem and phloem show variation in relative position in vascular bundles. The common arrangement is called **collateral** where xylem pipes are grouped towards the inner part of the stem and phloem takes the outer position (Figure 8)

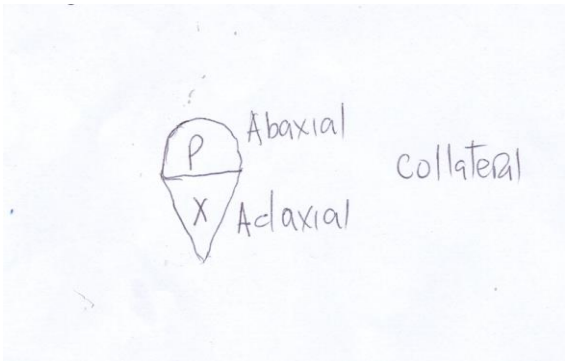


Figure 2.8: Vascular bundle showing common arrangement of xylem and phloem tissue

2.3 Leaves

In general sense there are quite a number of different types of leaves on a plant based on origin, structure and function. Based on these characteristics we come across four different types of leaves on a flowering plant i.e. cotyledons, cataphylls, floral and foliage leaves. However, when we talk of leaves we mostly refer to foliage leaves also known as true leaves.

2.3.1 Types of leaves and their functions

Cotyledons

Cotyledons are the first **embryonic** leaves on the plant. The main function of cotyledon is that they act as placenta and nurse the young plant.

Cataphylls

Structurally, you will notice cataphylls as bracts or scales on stems, roots and flowers of plants. They are found on either stolons, rhizomes of grasses or bulbs. Cataphylls offer protection to the shoot and they are also used for storage of food and nutrients.

Floral leaves

Floral leaves are modified for reproduction processes. Structurally, flowers are made up of four different types of leaves each modified for their own special functions. Such leaves include calyx, corolla, stamen and pistil.

Foliage leaves

As we have already indicated, these are the true leaves. This group of leaves represent the true leaves of the flowering plant. Structurally, the foliage leaf usually shows a specialization as a photosynthetic organ of the plant.

Major basic functions of foliage leaves include photosynthesis, gaseous exchange and translocation of manufactured food. However, they could also be modified to carry out other functions including storage, protection, propagation and holding onto other plants and objects.

2.3.2 External structure of foliage leaves

Foliage leaf normally shows its specialization as a photosynthetic organ by the expanded platform of its lamina (Figure 9). The leaf shows six major external parts including the tip/apex, base, blade/lamina, margins, main central vascular bundle (midrib) and the stalk (petiole and/or sheath). Though having leaves with these major parts, different plants show differences in size, shape, leaf arrangement and morphological appearance of their tips/apices, margins and bases. Leaves of

monocotyledonous and gymnospermae usually tend to have long and flat or round lamina while most leaves of dicotyledonous usually tend to be broad shaped.

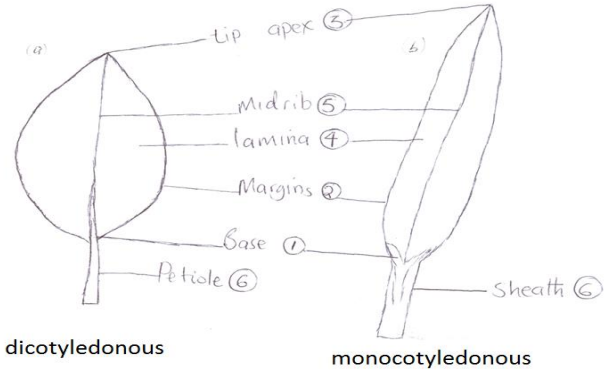


Figure 2.9: Typical dicotyledonous and monocotyledonous leaves

2.3.3 External variations of a foliage leaf

Below, we will study the common variations that we can see among foliage leaves of flowering plants. Apart from differences in shapes and size of the leaves (Figure 10), leaves show variation in tips, bases (Figure 11), margins and arrangement of vascular bundles (venation) (Figure 12).

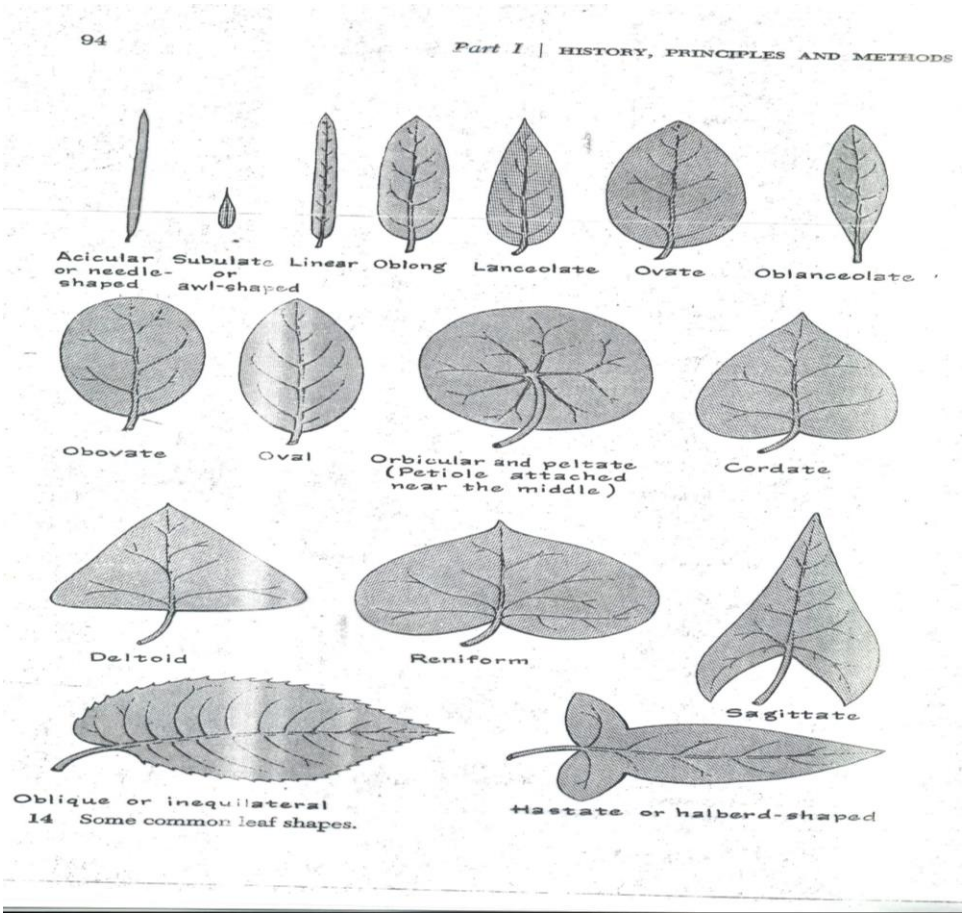


Figure 2.10 Common leaf shapes

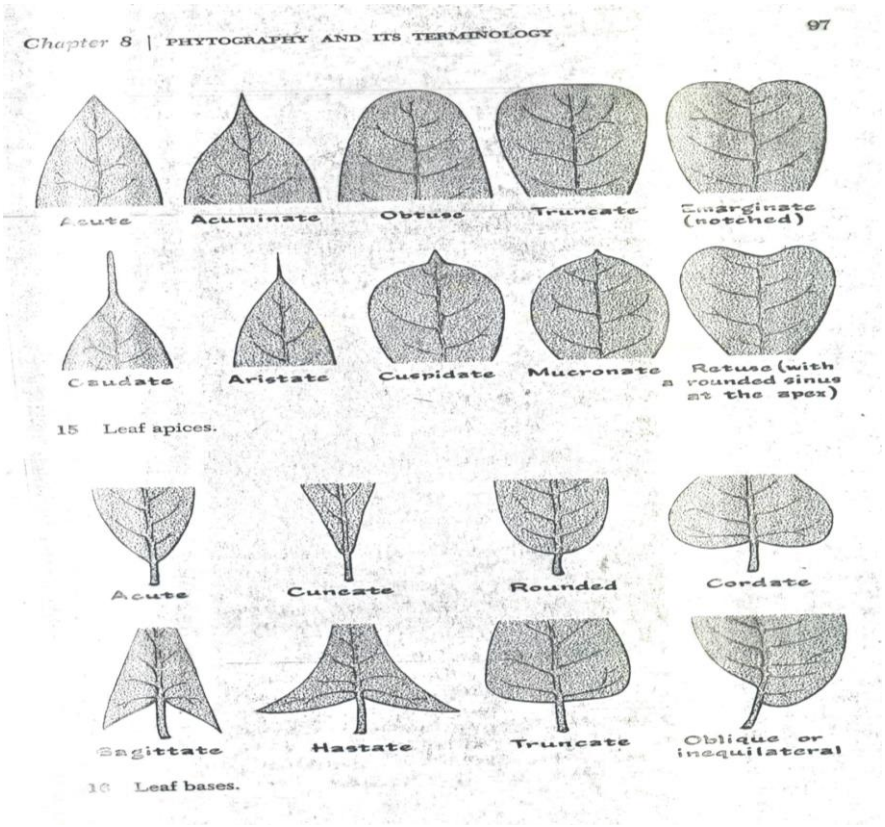


Figure 2.11 Common leaf apices and bases

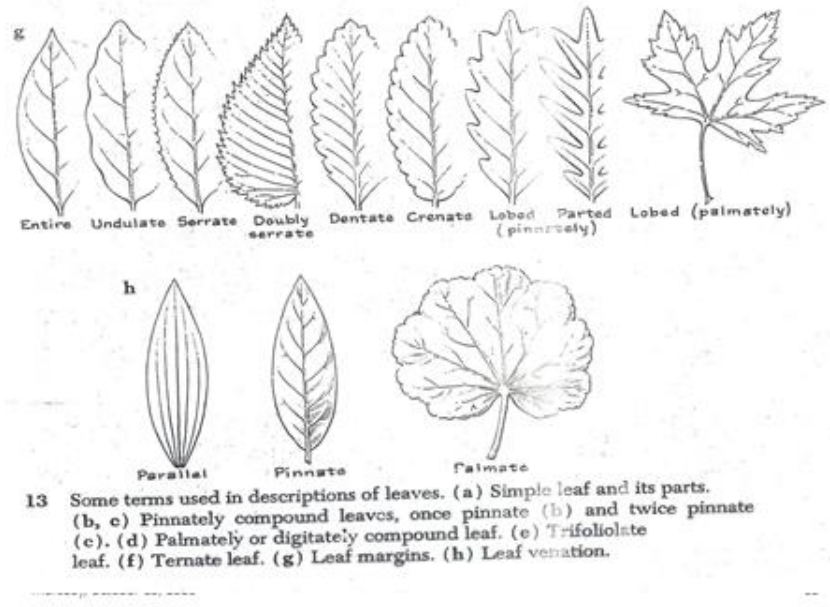


Figure 2.12 common leaf margins and arrangement of vascular bundles

Leaves may also vary in their arrangement on the stem, pattern of attachment on the stem and number of leaves attached at a node.

Leaf arrangement

This refers to the number of leaves on the node and how leaves position themselves in relation to one another. There are four common leaf arrangements. **Alternate leaf arrangement** where one leaf per node and these leaves are positioned in alternating positions for example maize and beans. In **Cyclic leaf arrangement** one leaf is attached per node but leaves are arranged in circular form going up the stem. **Opposite leaf arrangement** has two leaves per node positioned opposite to each other. The opposite arrangement could appear in two forms. Some plants carry their leaves on one plane others on two planes where two leaves on the lower node alternate with the two leaves immediately above them for example Gmalina. Lastly, **whorled leaf arrangement** where there are three or more leaves on one node and seem to make a whorl or ring around the stem.

Petiole versus Sessile leaves and sheathing base

In most plants, leaves are seen to be attached to leaf stalks as they immerge from a node. This stalk could be a strong round bodied structure (petiole) or flat flexible structure (sheath). Such leaf appearance could be regarded as a normal situation. However in some plats, leaves are seen to be directly emerging from the node. In such situations it is either the stalk is too short that it can be recognized or there is no stalk. In this kind of leaf attachment to the node, the leaf is referred to as a sessile leaf. In other cases, leaf blade coming from the sheath will also be seen coming straight from the stem with its sheath holding closely to the stem. Such appearance is referred to as a sheathing base. See diagrams below.

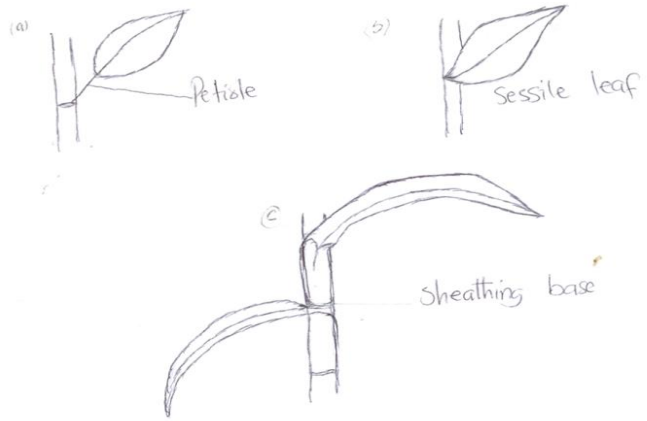


Figure 2.13: Leaf bases

Simple versus Compound leaves

A simple leaf has a single blade while the blade of a compound leaf is divided in various ways into leaflets (Figure 15). Regardless of the number of leaflets, a compound leaf still has a single axillary bud at its base. Pinnately compound leaves have the leaflets in pairs along an extension of the petiole called **rachis,** while palmately compound leaves have all the leaflets attached at the same point at the end of the petiole. Sometimes, the leaflets of a pinnately compound leaf may be subdivided into still smaller leaflets forming a bipinnately compound leaf and a further subdivision forming a tripinnately compound leaf (Figure 16).

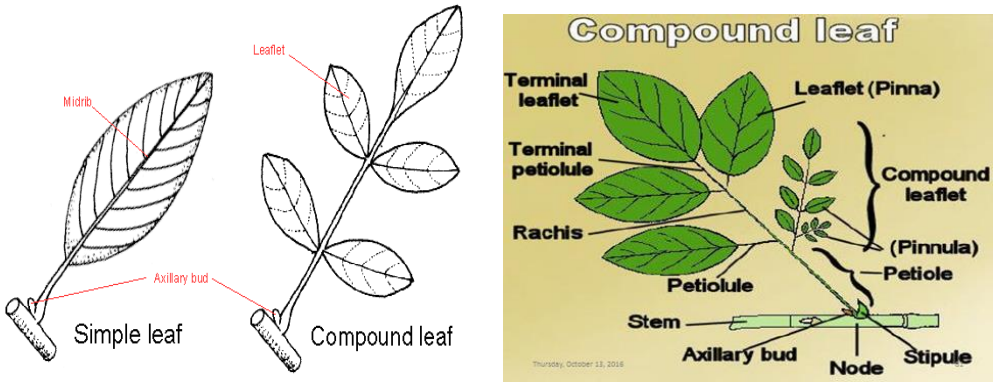


Figure 2.14 Simple and compound leaves

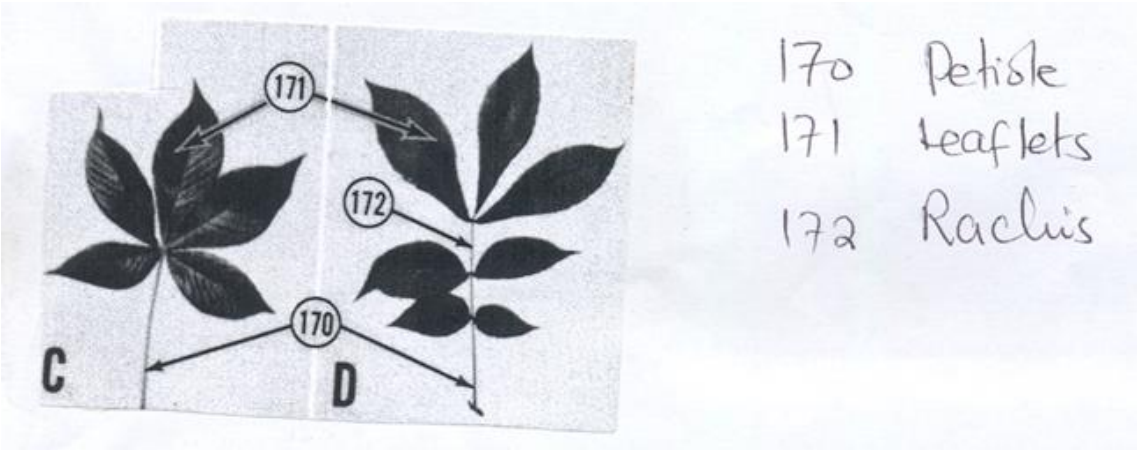


Figure 2.15: Palmately compound leaf (C) and pinnately compound leaf (D)

2.3.4 Internal structure of a leaf

Internal structure of leaves vary depending on the prevailing environmental conditions of the habitat plants have been adapted to. Plants can be grouped into three groups based on their habitat and internal structural leaf arrangement. Such groups include mesophytes, hydrophytes and xerophytes. **Mesophytes** are plants that require environment that is neither too wet nor too dry. Hydrophytes plants require a large supply of water and may grow partly or entirely submerged in water while **Xerophytes** are adapted to a dry hot habitat.

Internal structure of a mesophytic leaf

Like roots and stems, the leaf also shows three major groups of tissues namely dermal vascular and ground tissues (Figure 17).

Epidermis

The leaf is covered/protected by two skins, with both made up of similar type of cells (epidermal cells) the skin that appears on the upper side of the leaf is referred to as an upper epidermis and the one below is referred to as lower epidermis. For its efficient functional purposes, epidermis is equipped with some structures including cuticle, trichomes and stomata.

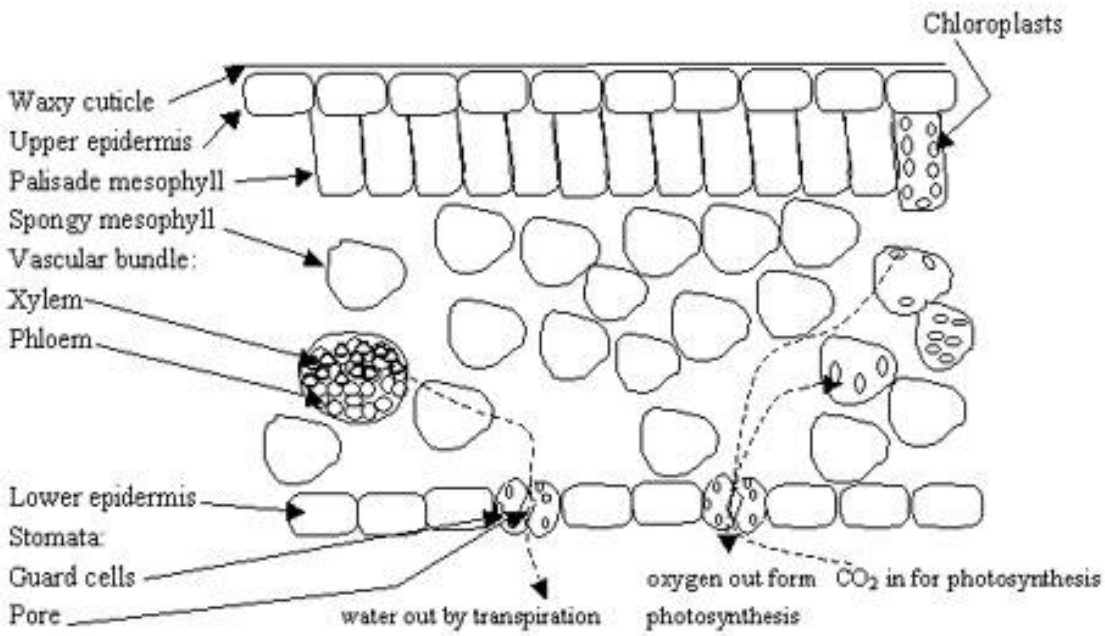


Figure 2.16: A composites cross section of a mesophytic leaf

Cuticle is a fatty waxy substance which repels water. It is mainly coated on the surfaces of the epidermis for the same purpose of preventing direct entry of water from the atmosphere. The other common structures found on the epidermis are trichomes (hairs). The epidermal layer of roots has root hairs for the absorption of

water. The hairs either on stems or leaves are mainly for protection and temperature regulation. The other popular structures found either on both or one side are the natural openings called **stomata**. These act as openings for gaseous exchange. Based on the position of the stomata, leaves may be referred to **Amphistomatic** leaf if stomata appear on both sides. Epistomatic have stomata on the upper side while hypostomatic have stomata on the lower side.

Ground tissue

The main part of the body of the leaf is covered by ground tissue and in leaves is referred to as **mesophyl**. The mesophyll has the following characteristics:

a. mainly composed of parenchyma cells of varying sizes and shapes
b. cells contain many chloroplasts
c. large volume of intercellular spaces
d. It could be homogeneously be divided into palisade Mesophyll with elongated cells appearing in columns or Spongy Mesophyll with regular sloped cells.

Leaves may have one or more rows of palisade parenchyma cells. Leaves of plants adapted to mesophytic habitats commonly have one row of palisade parenchyma cells. On the other hand leaves adapted to xerophytic habitat tend to have two or more rows of palisade parenchyma cells.

Vascular tissue

The Vascular System of the leaf is distributed throughout the blade and thus shows a close spatial relation to mesophyll. Vascular strands form an interconnected system in the median plane of the blade parallel with the surface of the leaf. The Vascular bundles in the leaf are commonly called **veins**, and the patterns formed by the veins, **venation**, as seen with the unaided eye. The venation appears in two main patterns, the **reticulate** or **netted** which is mainly associated with leaves of

dicotyledonous plants**,** and the **parallel** which is mainly associated with leaves of monocotyledonous plants.

2.4 Flowers

2.4.1 Functions, structure and variations.

You will recall that it was mentioned that flowers are a form of modified leaves. They are modified for a purpose of carrying out reproductive processes. Structurally, a standard flower depicts four groups of such modified leaves but this is not common to all flowers. Other structural variations will be observed in size, colour, number of parts and arrangement of these floral parts. Below we intend to discuss functions, structure and common variations in flowers.

A flower as a group of specialized floral leaves are adapted for: -

a) Production and development of reproductive cells (pollen ♂ and ovules ♀)
b) Pollination:- the transfer of pollen
c) Fertilization:- the unification of two protoplasmic bodies usually from different individual plants to form one body (**protoplast - zygote**).
d) The ultimate development of the seed and fruit.
e) Successful dispersal of the seeds (fruits)

2.4.2 Morphological structure.

Flowers are of many different forms. They also vary greatly in size, colour, number of parts and arrangement of parts. Each flower is an assemblage of fertile and sterile organs in preparation for the five activities above. See figure 18 below.

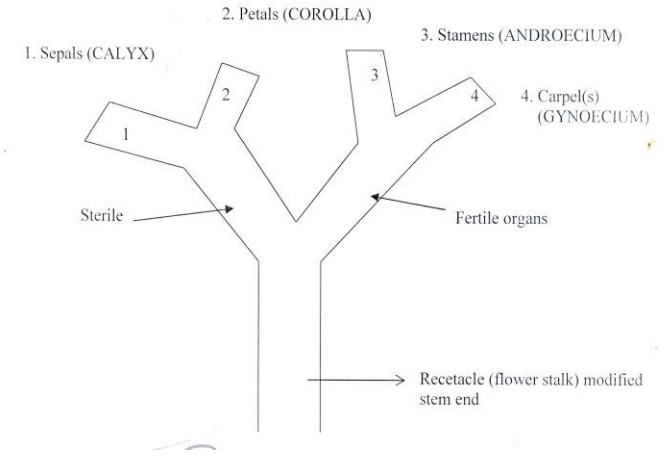


Figure 2.17: Classification of floral parts

A typical flower is composed of four whorls of modified leaves. These are sepals, petals, stamens, and a carpel or carpels. All attached to the receptacle, the modified stem end that supports these structures (Figure 19).

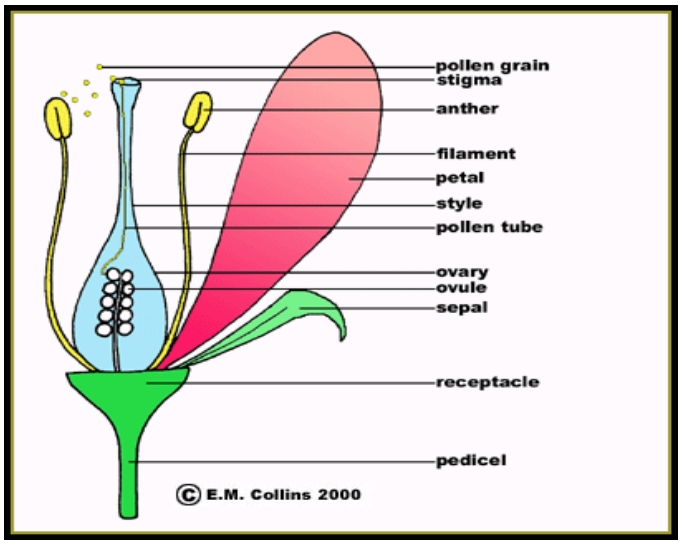


Figure 2.18 a typical flower

The **sepals** enclose the other flower parts in the bud. Generally, they are green. All the sepals taken collectively constitute the calyx; that is, the catalyx of a flower is composed of more or less distinct parts, the sepals.

The **petals** are usually the conscpicuous, colored, attractive flower parts. Taken together, the petals constitute the corolla.

The **stamens** form a whorl, lying inside of the corolla. Each stamen has a slender stalk or filament, at the top of which is an anther, the pollen-bearing organ. The whorl or grouping of stamens is called the **androecium**. Collectively the carpels are spoken of as the **gynoecium**. Each individual structure in the gynoecium is commonly referred to as a pistil. There are generally three distinct parts to each pistil.

a) An expanded basal portion, the ovary, in which are borne the ovules,
b) The style a slender hollow stalk which lies between the ovary and the stigma above.
c) The stigma: The pollen is deposited on the stigma for initiation of successful fertilization.

Summary on the floral parts

The parts of the flower in outline are as follows:

Receptacle

Calyx, consisting of sepals
Corolla, consisting of petals

sterile parts collectively known as perianth

Androecium, consisting of stamens
Gynoecium, consisting of carpels

fertile and essential parts

Variations

(a) **complete and incomplete / perfect vs imperfect**

If a flower has a stamen, pistil, petals, and sepals, it is called a **complete** flower. If one of these parts is missing, the flower is called **incomplete**. The stamen and pistil are the essential parts of a flower and are involved in seed production. If a flower contains both functional stamens and pistils, it is called a **perfect** flower, even if it does not contain petals and/or sepals. If either stamens or pistils are lacking, the flower is called **imperfect**. Female (**Pistillate)**, and Male (**Staminate**) flowers (figure 18).

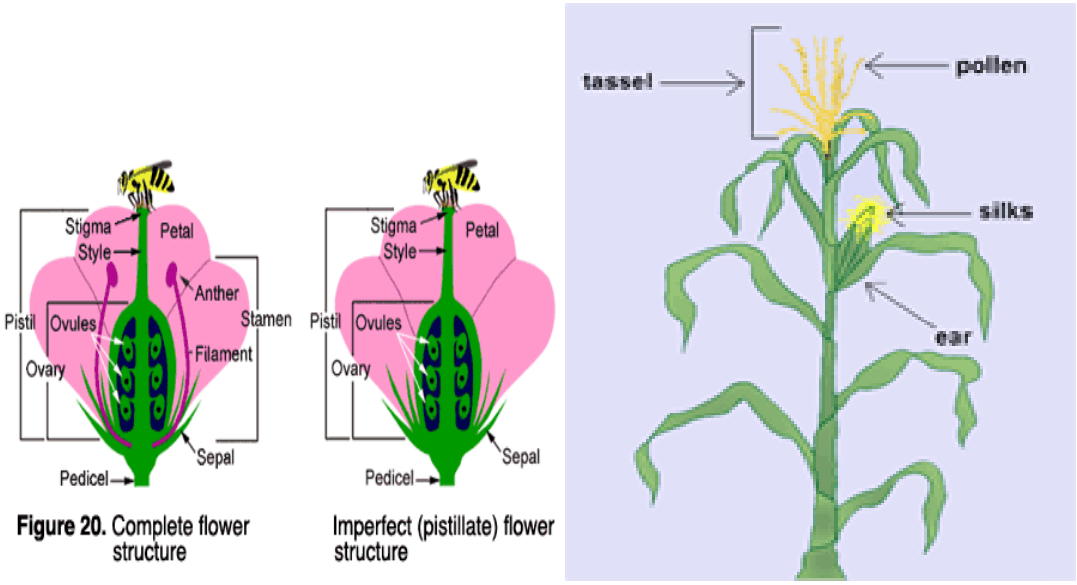
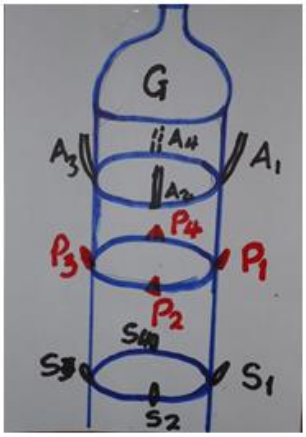


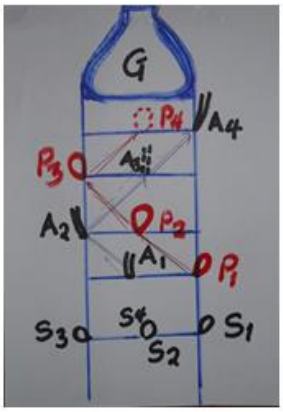
Figure 2.19: Complete and incomplete flowers

(b) Whorled and spiral arrangements

In most flowers the sepals, petals, stamens, and carpels are in whorls or circles on the axis of the flower. In contrast with this cyclic or whorled arrangement of flower parts is the arrangement in which one or more sets of flower parts are in spirals.



x (whorled/cyclic)



y (spiral)

G = gymnoecium
A = androecium
P = petals
S = sepals

Figure 2.20: Whorled versus spiral arrangement

(c) Union of flower parts. *coalescence*.

In many flowers, the members of one or more whorls are to some degree united with one another, or are attached to members of other whorls. The union with each other of members of a given whorl is termed **coalescence.**

Whorl	**No Coalescence**	**Coalescence**
Sepals	aposepalous	synsepalous, synsepaly
Petals	apopetalous	sympetalous, sympetaly
Stamens	apoandrous	synandrous, synandry
Carpels	apocarpous	syncarpous, syncarpy

(d) Elevation of flower parts

In this variation, we consider the relationship between the position of the ovary and the rest of the other parts. In some flowers, the receptacle is convex or conical and the different flower parts are arranged one above the other. They occur in the following order (beginning with the lowest): sepals, petals, stamens, and carpels. A flower of this type is said to be **Hypogynous**. (Superior Ovary) (Figure 20).

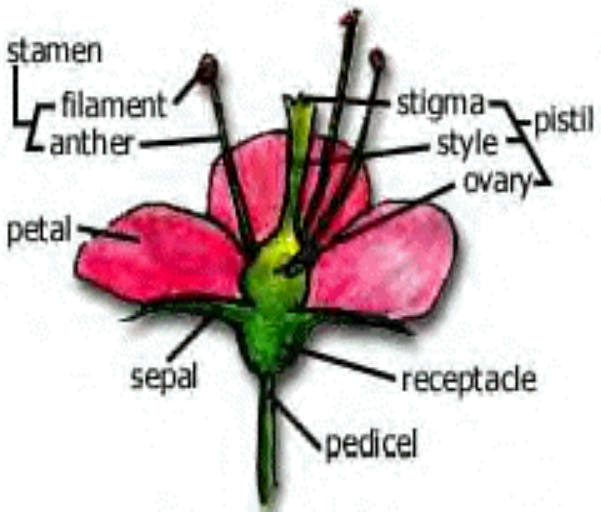


Figure 2.21 superior ovary positioned above other floral parts

If the sepals, petals and stamens appear to come from the top of the ovary, the flower is said to be **Epigynous** (Inferior Ovary) (figure 21). Intermediate between the hypogynous and **epigynous** types are those flowers in which the sepals, petals and stamens have their origin, not above or below the pistil, but around the ovary. This type of flower is said to be **perigynous** (Figure 22).

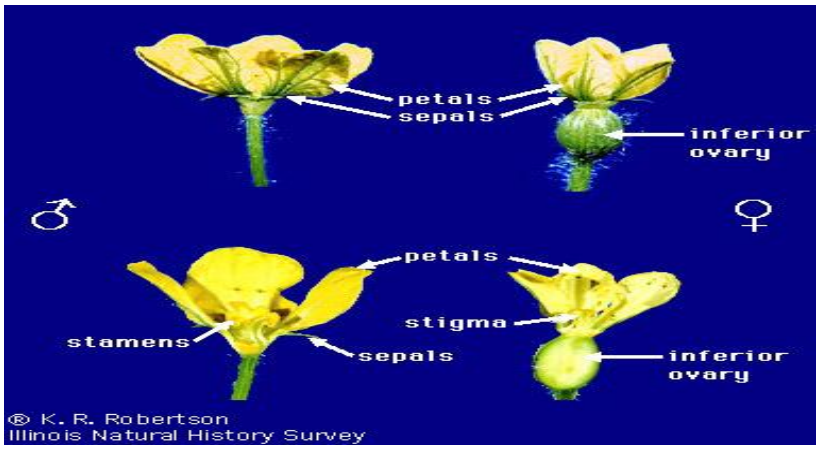


Figure 2.22: inferior ovary epigynous as depicted by pumkin flowers

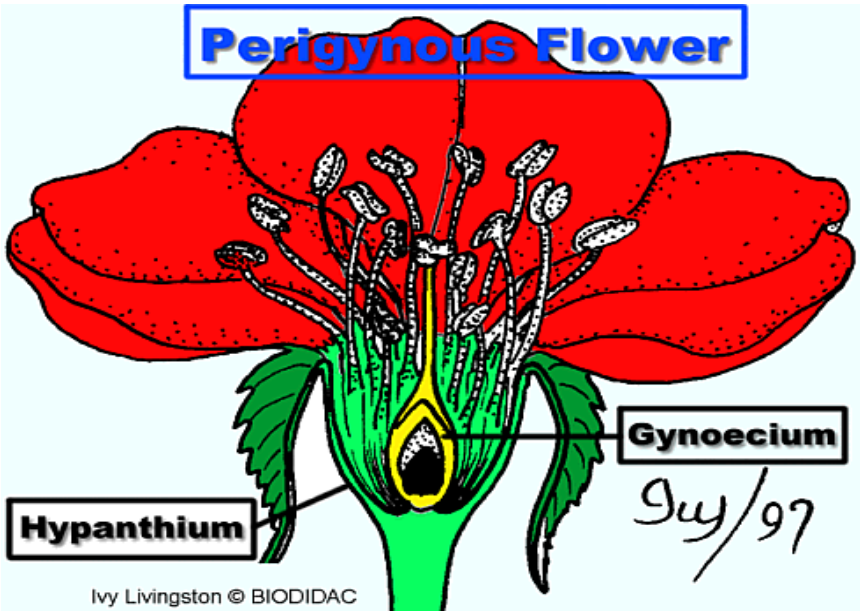


Figure 2.23: intermediate ovary perigynous

(e) The composite flower

As the name implies the composite flower is a group of many small flowers arranged to give the appearance of a single typical flower. Good examples are depicted by sunflower, pineapples, mangoes and maize (Figure 21).

Figure 2.24 Composite flowers

(f) The Grass Flower

Flowers of monocotyledonous plants have their flower parts (whorls), in threes or multiples of three as opposed to flowers of dicotyledonous plants which tend to be in fours or fives or multiples of four and/or five i.e. sepals, petals, stamens, and gynoecium.

The grass flower is regarded as an extremely reduced form of a basic monocotyledonous flower type. In the gramineae family (Grasses), the **perianth** (Sepals + Petals), is reduced to two segments which appear as scales called **lodicules**. Stamens appear in threes with very large lobes of anthers (Two on each filament). Gynaecium is reduced to a single chamber ovary (One seed ovary) with two feathery styles and with a wide feathery stigma. Such flower is referred to **floret** (Figure 24). The flower is adapted to wind pollination.

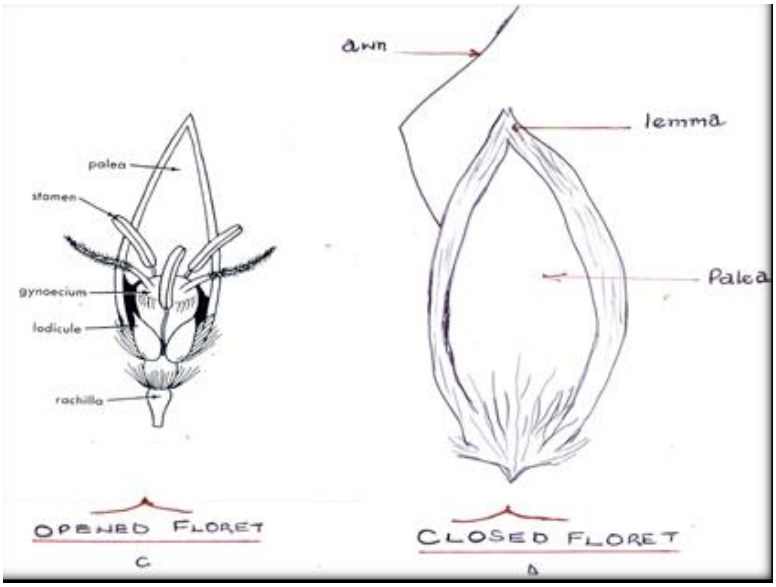


Figure 2.25 Individual grass flower called floret

Individual flower in grasses appear as composite flowers in form of inflorescence (Figure 25). Each ellipse represents a complete spikelet. Grass flowers generally grow in a head or spike.

In all instances the individual small flowers or florets are associated in groups known as **spikelets.** Each spikelet is separated from its neighbours by two small modified leaves or bracts, known as **glumes** (Figure 26). They are found at the base of the spikelet. A separation of the glumes reveals the individual florets which vary in numbers ranging from three to eight depending on type grasses. The spikelet is attached to a slender stalk called the **rachilla**. Each floret is in turn is protected by two additional bracts, the **lemna** and the **palea** (Figures 24 and 26).

The lemna may have attached to it a long slender whip-like structure called **awn**. The essential parts consist of the **androecium** composed of three stamens and the **gynoecium** of **two fused carpels.** When the flowers are mature the palea and lemna separate slightly, exposing two featherly stigmas and allowing the anthers

to hang free from the spikelet on greatly lengthened filaments. At the base of the floral parts there may be distinguished two small protuberances called **lodicules**. These are thought to be the greatly modified perianth parts and are supposed to function in the separation of the lemna and palea at pollination time. Lodicules act as sponges, when filled with water they expand hence pushing lemna and palea apart.

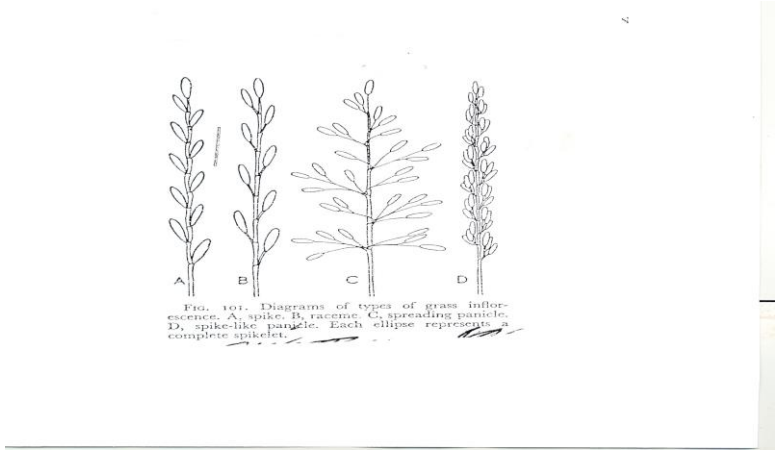


Figure 2.26: Common types of grass inflorescence

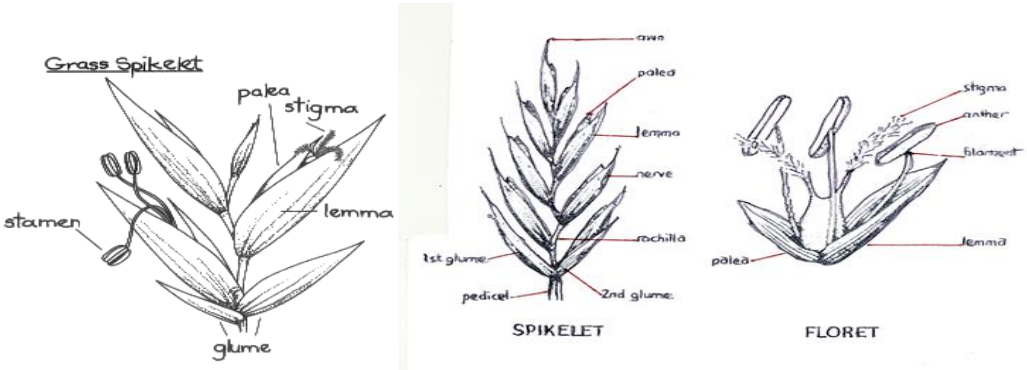


Figure 2.27 Spikelet and floret

Unit 2 Practical activities

Refer to Unit 2 of the practical module.

Unit 2 Summary

In this unit, we have discussed the morphology, anatomy and functions of the basic organs that make a flowering plant ie roots, stems, leaves and flowers.

Roots are basically there to provide anchorage of the plants in the soil. However, they also have other functions including absorption and transportation of water and mineral salts from the soil, storage of manufactured food, translocation of manufactured food to other required places, and other specialised functions. Roots are basically devided in three structural and functional groups called tissues. Such tissues include dermal, ground and vascular.

Stems are subdivided into nodes and internodes. Leaves and any other appendages of the plants grow from the nodes of the stems. Apart from this function, stems are also known to transport water from roots to leaves. They also translocate manufactured food between leaves and roots and within its own body. Other functions include storage, support and other specialised ones as discussed in the text. Like in roots, they also show three structural and functional groups of tissues are present in the stem. However the arrangement of these tissues especially the ground and vascular is different in stems as in roots.

We have learnt that based on origin, structure, function and position, plants may produce four different types of leaves i.e cotyledons, cataphylls, floral and foliage leaves. Much emphasis of our discussion has been mainly on the foliage leaves. These leaves are basically present on the plants for

photosynthetic activities. Apart from this function they also perform other specialised functions. As a photosynthetic apparatus, leaves tend to be broad-shaped as solar panels. The leaf body is covered with two skins, one on the upper side and another on the lower side. The space between the two skins is mostly covered by ground tissues. But here and there, we see vascular tissues in form of veins appearing.

Flowers in plants appear as modified stems and leaves for reproductive purposes. Each standard flower is made up of four basic parts i.e. sepals, petals, stamens and pistils. However, we see a lot of variations in flowers with regards to the numbers, arrangement, sizes, colour and shapes of these four parts.

Unit 2 Test

1. Name the three basic tissues present in all three vegetative organs of a plant.
2. With aid of well labelled diagrams, show how the above tissues are arranged in:
   a) Roots
   b) Monocotyledonous stems
   c) Dicotyledonous stems
   d) Leaves
3. Draw and label a root tip
4. State five main functions of a normal root
5. Name and state a function of any five specialised roots
6. Define the following terms:
   a) Meristem
   b) Sessile leaf
   c) Cataphylls
   d) Senescence

e) Floret

f) Perfect flower

g) Trichome

h) Whorled arrangement of leaves

i) Epidermis

j) Phloem

7. Differentiate between:

a) Taproot and adventitious root systems

b) Arrangement of vascular tissues in dicotyledonous and monocotyledonous stems

c) Unisexual and bisexual flowers

d) Amphistomatic and hypostomatic leaves

e) Annual and perennial plants

f) Hypogenous and epigynous flowers

8. Write short notes on the following:

a) Veination in plant leaves

b) Patterns of leaf arrangements on the stem

c) Life cycle of flowering plants

d) Endodermal tissue of the root

Unit 2 answers

Reading Assignment

Unit 3: Introduction to genetics

Introduction

You may be surprised as you read human history that people were unaware of the scientific details of how offspring were conceived and how heredity worked. Clearly they noticed the hereditary relationship between parents and children but not hereditary mechanisms.

In this unit we will try to understand the genetic basis of hereditary mechanism. However, before we discuss these we will start by recapping what you learnt in Biology I module, unit 2; structure and function of the cell. We will also look at the theories that tried to explain the origin of life and how variations occur between parents and offspring. Finally we will discuss how allele and genotype frequencies vary from generation to generation within a family and populations.

Unit 3: learning objectives

By the end of this unit, you should be able to:-

a. recap the structure and function of a cell
b. discuss the theories of the origin of life
c. evaluate Mendelian monohybrid ratios, dihybrid ratios and test cross
d. identify conditions that may lead to a modification of Mendelian monohybrid and dihybrid ratios
e. analyse allele and genotype frequencies in a population

Unit 3 key terms

Ensure that you understand the key terms or phrases used in this unit as listed below:-

chromatin

nucleolus spontaneous generation

inheritance of acquired characteristics

blending theory

particulate theory

cross-fertilisation

self-pollination

hybridisation

monohybrid

F1 generation

F2 generation

law of segregation

law of independent assortment

test cross

heterozygosity

zygosity

dominance

Unit 3 Content

3.1 Structure and function of the cell (recap)

By now you should know that although living things have different forms, they are all cellular meaning that they are composed of at least one cell as a basic unit of life and that each cell comes from a pre-existing cell. Robert Hooke was the first to describe cells after observing cork slices using a microscope and coined the term Cell.

You will also appreciate that our knowledge of cells has grown dependent on advances done in the field of microscopy.

In unit 2 of Biology I, we discussed that cell at its basic structure has cell membrane, nucleus, organelles and the cytoplasm. The **cell membrane** is a thin bi lipid layer which separates the cell contents from its immediate environment. Plant cells also have in addition the cell wall which gives shape to the cell. The **nucleus** is specialised for storage of DNA and controls cell functioning including reproduction. The **cytoplasm** is a liquid substance between the nucleus and the cell membrane in which the organelles are located. **Organelles** on the other hand are small bodies with specific structures and functions within the cell. The organelles include the endoplasmic reticulum (both smooth and rough), ribosomes, polysomes, golgi apparatus, vacuoles and storage vesicles, mitochondria, chloroplasts and plastids, cytoskeleton, microtubules and microfilaments. Some cells may contain cilia and flagella. In addition plants have chloroplasts and plastids for harvesting light energy for photosynthesis.

Our interest in this section is to revise structure and function of the nucleus. The nucleus has two major functions: it stores the cell's hereditary material and it coordinates the cell's activities. Do not forget that only eukaryotic cells have a nucleus enclosed in a nuclear membrane

The nucleus is membrane bound enclosing a semifluid matrix called nucleoplasm. The nucleoplasm contain most of the cells genetic materials organized as long multiple linear DNA molecules in complex with a large variety of proteins such as histones to form chromosomes. Chromatin is the less condensed form of the nucleus. The DNA is the genetic code of the cell.

The nucleus also contains one or more nucleoli. Nucleoli are nucleus organelles that synthesize protein-producing macromolecular assemblies called ribosomes and a variety of other smaller components. Through the microscope, the nucleolus looks like a large dark spot within the nucleus. A nucleus may contain up to four nucleoli, but within each species the number of nucleoli is fixed. After a cell divides,

a nucleolus is formed when chromosomes are brought together into nucleolar organizing regions. During cell division, the nucleolus disappears.

A section of a chromosome which controls a particular trait is a gene. A locus (plural: loci) is the position of a gene (Figure 3.1). The genes within these chromosomes are the cells nuclear genome and structured in such a way to promote cell function. The gene controls cellular activities by production of proteins. Other places where you find DNA molecules are mitochondrion and chloroplasts. DNA in these organelles are involved in the control of the specific processes of those organelle.

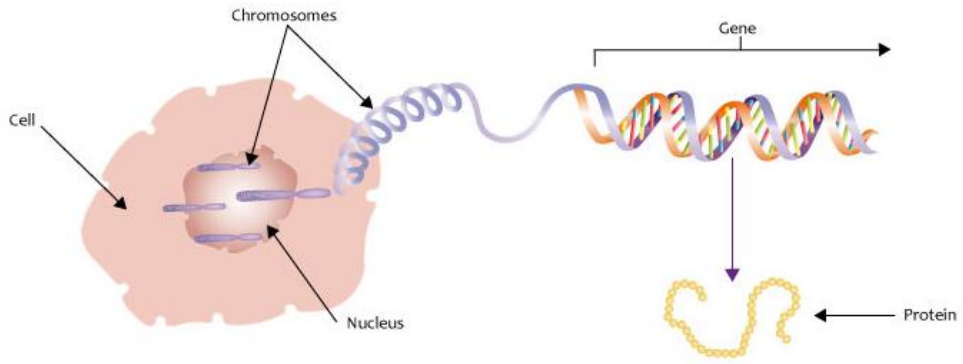


Figure 3.1: The central dogma of molecular biology

At this point I am sure you are not surprised why the nuclear envelope is a double-layered membrane with space between the layers called the perinuclear space and appears to connect with the rough endoplasmic reticulum. The envelope is perforated with tiny holes called nuclear pores. These pores regulate the passage of molecules between the nucleus and cytoplasm, permitting some to pass through the membrane, but not others. The inner surface has a protein lining called the nuclear lamina, which binds to chromatin and other nuclear components. During mitosis or cell division, the nuclear envelope disintegrates but reforms as the two cells complete their formation and the chromatin begins to unravel and disperse.

3.2 Theories of the origin of life

Together with the advancement in knowledge about the cell did the motivation to answer the question about the origin of life (the cell) advanced. Scientists have tried to explain the origin of life on earth which have evolved to produce a number of theories.

3.2.1 Spontaneous generation

This is the first theory to be recorded that tried to address the origin of life. The belief in spontaneous generation where life is believed to have evolved from non-living matter dates back to the time of Aristotle and ancient Greek philosophy. The theory was advanced together with a belief in heterogenesis where one form of life can be derived from a different form.

The first experimental evidence against spontaneous generation came in 1688 when Francesco Redi showed that no maggots appeared in meat when flies were prevented from laying eggs. In 1768, Lazzaro Spallanzani demonstrated that microbes were present in the air and could be killed by boiling. In 161, Louis Pasteur performed a series of experiments that demonstrated that organisms such as bacteria and fungi do not spontaneously appear in sterile, nutrient rich media but could only appear by invasion from without.

As you can see, the belief of spontaneous generation was disapproved and begged for an alternative theory. Besides the theory also failed to explain the occurrence of variations among parents and offspring.

3.2.2 Inheritance of acquired characteristics

This theory was developed to partially address the existence of variations among parents and offspring. The theory was developed by Lamarck who argued that physiological changes acquired over the life of an organism (such as the enlargement of a muscle through repeated use) may be transmitted to offspring. It is also commonly referred to as the Theory of Adaptation which is related to the Evolutionary Theory of French naturalist Jean-Baptiste Lamarck known as Lamarckism.

This theory was disapproved by a simple interesting experiment conducted by Weismann who removed the tails of 68 white mice, repeatedly over 5 generations and observed that no mice were born in consequence without a tail or even with a shorter tail. Unfortunately, the theory was consequently disproved and that necessitated another theory.

3.2.3 Blending theory

Weismann then developed the blending theory which aimed at explaining the existence of variations between parents and offspring. The theory states that offspring express phenotypic characters intermediate between those of the parents. However, the theory could not explain how traits that seemingly disappeared for several generations often reasserted themselves down the line unaltered. The theory could not explain for example the existence of some characteristics such as albinism from parents with normal skin and hair colour.

3.2.4 Mendelian inheritance theory

Gregor Mendel (1822-1884) was an Austrian monk who discovered the basic rules of inheritance. From 1858 to 1866, he bred garden peas, *Pisum sativum*, in his monastery garden and analysed the offspring of these crossings.

Gregor Mendel proposed Inheritance theory/ Particulate theory which states that similarities and differences between parents and off springs are as a result of passage of discrete hereditary factors/genes. The work of Mendel was rediscovered after many years after his work was rejected theory when they could not find a robust theory to address the shortfalls in the previous theories.

Mendel succeeded to come up with principles that lead to the development of this theory that has not been challenged to date because of the following factors:-

a) knowledge of statistics which he used to predict and verify outcome of his experiments
b) choice of subjects for experiments, he used garden peas which can be cross pollinated or self-pollinated
c) choice of traits, he looked at one trait at a time
d) many varieties were available that bred true for clear cut quantitative traits analysis such as seed texture, seed colour, flower colour, growth habit
e) the way of presenting results, he used numbers to represent whatever he discovered
f) his background, he was interested in breeding experiments

3.3 Mendelian monohybrid, dihybrid ratios and test cross

Prior to his work with pea plants plant breeders had conducted experiments aimed at obtaining flowers with new varieties of colours. You probably are aware that when two distinct individuals with different characteristics are crossed to each (hybridization) their offspring are referred to as hybrids. Mendel chose the garden pea to investigate the natural laws that govern plant hybrids. In some experiments, Mendel carried out self-fertilisation which means that the pollen and egg were derived from the same plant. Cross pollination is when pollen from one plant is placed on a stigma of a different plant.

Before starting his experiments he concentrated on obtaining several varieties of peas that were considered to be distinct. These plants were different with regards to many morphological characteristics. The general characteristics of an organism are called characters. The term trait and variant are typically used to describe the specific properties of a character. A variety that produces the same trait after several generations of self-fertilization is called true breeding line.

After obtaining true breeding lines for seven traits (Figure 3.2), Mendel studies these characters by crossing the variants to each other. A cross in which an experimenter is observing only one character is called monohybrid cross also called single factor cross. When the two parents are different variants for a given character, this type of cross produces single character hybrids also called monohybrids.

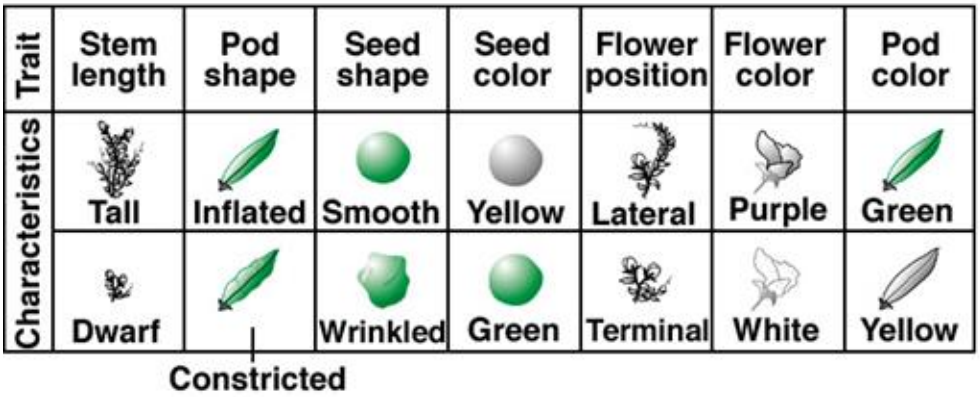
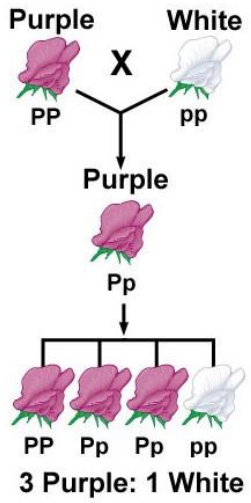


Figure 3.5: Seven Characteristics of peas Mendel studied

Mendel's experimental procedure began with true breeding plants that differed with regards to a single character. These are termed the parental generation of P generation. When the true breeding parents were crossed to each other, this is called p cross and the offspring constitute the F1 generation for first filial generation. Mendel observed that all F1 generation showed the phenotype of one parent. This prompted Mendel to follow the transmission of this character

for another generation. To do this, the F1 generation were allowed to self-fertilize to produce a second generation called F2 generation for second filial generation.



P = Parentals (True breeding, Homozygous and Homogametic)

F1 generation (Heterozygous, Heterogametic)

F2 generation (Gives 3:1 Mendelian monohybrid ratio)

Figure 3.3: Mendelian monohybrid cross

From the data that he obtained from this experiment, Mendel made a quantitative analysis of data from which he was able to propose the following important ideas.

a) Mendel data argued strongly against a blending mechanism of heredity.
b) Variant for one character is dominant over another variant. The term recessive is used to describe a variant that is masked by the presence of a dominant trait that is masked by the presence of a dominant trait but reappears in subsequent generations.
c) Dominant trait was always observed in F1 generation for heterozygous plants.
d) Genetic determinants of traits are passed along as unit factors from generation to generation. His data were consistent with a particulate theory

of inheritance in which the genes that govern traits are inherited as discrete units that remain unchanged as they are passed from parents to offspring.

e) Mendel compared the numbers of dominant and recessive traits in the F2 generation, he noticed as recurring pattern. Within experimental variations, he always observed approximately a 3:1 ratio between the dominant traits and the recessive trait.

Mendel was the first scientist to apply this type of quantitative analysis in a biological experiment. This quantitative approach allowed him to propose a law of segregation of genes.

Law of Segregation of genes (the "First Law")

The Law of Segregation states that every individual organism contains two alleles for each trait, and that these alleles segregate (separate) during meiosis such that each gamete contains only one of the alleles. An offspring thus receives a pair of alleles for a trait by inheriting homologous chromosomes from the parent organisms: one allele for each trait from each parent.

Molecular proof of this principle was subsequently found through observation of meiosis by two scientists independently, the German botanist Oscar Hertwig in 1876, and the Belgian zoologist Edouard Van Beneden in 1883. Paternal and maternal chromosomes get separated in meiosis and the alleles with the traits of a character are segregated into two different gametes. Each parent contributes a single gamete, and thus a single, randomly successful allele copy to their offspring and fertilization.

Law of Independent Assortment (the "Second Law")

After understanding heredity mechanisms of monohybrid traits Mendel proceeded to investigate the pattern of inheritance for two different characters. He carried out two factor crosses also known as dihybrid crosses in which he

followed the inheritance of two different characters within the same groups of individuals.

Let us consider an experiment in which one of the characters was seed shape found in round or wrinkled variants, the second character was seed colour which existed as yellow and green variants.

Mendel took peas with round (R) and green (Y) seeds and crossed them with wrinkled (R) and yellow (y) seeds in a cross as shown below.

P1:	RRYY	x	rryy
Gametes (heterogametic):	RY		ry
F1:	RrYy (round and green seeds)		

He then selfed F1 to obtain F2 and or this we use a Punnett square as shown in Figure 3.4 below.

	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

Figure 3.4: Punnet square for Mendelian dihybrid cross

This gave Mendelian dihybrid ratio, 9:3:3:1 of round green seeds: round yellow seeds: wrinkled green seeds: wrinkled yellow seeds. From these results, he came up with the law of independent assortment which states that alleles for

separate traits are passed independently of one another from parents to offspring.

You can simplify this when you consider that the law states that the biological selection of an allele for one trait has nothing to do with the selection of an allele for any other trait. Mendel found support for this law in his dihybrid cross experiments. In his monohybrid crosses he idealized a 3:1 ratio between dominant and recessive phenotypes while in dihybrid crosses he found a 9:3:3:1 ratio. This shows that each of the two alleles is inherited independently from the other, with a 3:1 phenotypic ratio for each.

Test cross

In genetics, a test cross, first introduced by Gregor Mendel, involves the breeding of an individual with a phenotypically recessive individual, in order to determine the zygosity of the former by analysing proportions of offspring phenotypes. Zygosity can either be heterozygous or homozygous. Those that are heterozygous have one dominant and one recessive allele. Individuals that are homozygous dominant have two dominant alleles, and those that are homozygous recessive have two recessive alleles

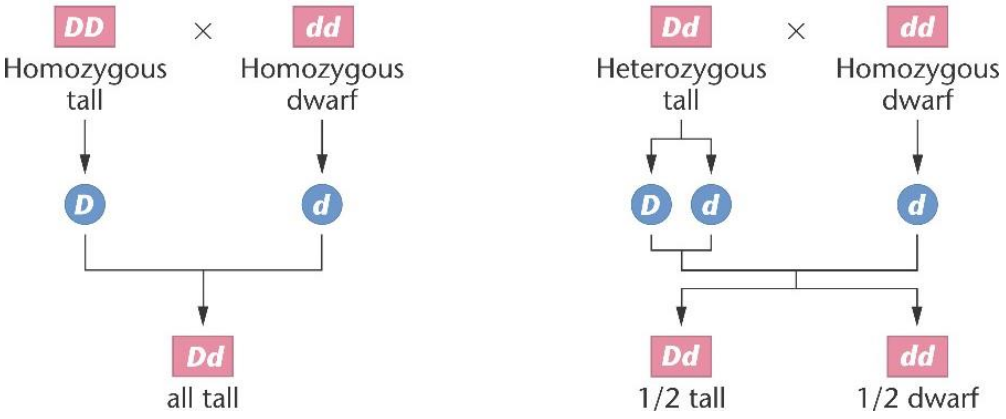


Figure 3.5: Genetic test cross

Test crosses involve breeding the individual in question with another individual that expresses a recessive version of the same trait. Analysing the proportions of dominant and recessive offspring determines if the individual in question is homozygous dominant or heterozygous. If all offspring from the test cross display the dominant phenotype, the individual in question is homozygous dominant; if half the offspring display dominant phenotypes and half display recessive phenotypes, then the individual is heterozygous (Figure 3.5). Since the homozygous recessive individual can only pass on recessive alleles, the alleles the individual in question passes on determine the phenotypes of the offspring.

Modification of Mendelian monohybrid and dihybrid ratios

Mendel explained inheritance in terms of discrete factors ‘genes’ that are passed along from generation to generation according to the rules of probability. Mendel's laws are valid for all sexually reproducing organisms. However, Mendel's laws stop short of explaining some patterns of genetic inheritance. Cases where Mendel's laws can strictly account for the patterns of inheritance are relatively rare. Often, the inheritance patterns are more complex.

Until the end of this section we will discuss some of the cases which may lead to modification of Mendelian monohybrid and dihybrid ratios which include codominance, multiple alleles, lethal alleles, duplicate genes, epistasis, non-epistasis, complementary genes, polygenic, linkage, sex linkage, autosomal linkage and mutations,

Dominance is a relationship between alleles of one gene in which the effect on phenotype of one allele masks the contribution of a second allele at the same locus. The first allele is dominant and the second allele is recessive. Dominance is a key concept in Mendelian inheritance and classical genetics.

Often the dominant allele codes for a functional protein whereas the recessive allele does not. Inheritance of such alleles reads to the classical Monohybrid ratio of 3:1 or Mendelian dihybrid 9:3:3:1 ratio.

Incomplete dominance

Incomplete dominance is a form of intermediate inheritance in which one allele for a specific trait is not completely expressed over its paired allele. This results in a third phenotype in which the expressed physical trait is a combination of the phenotypes of both alleles.

e.g.	P1 Flower colour:	RR (red)		x	WW (white)
	Gametes (Homogametic)		R		W
	F1:		RW (pink)		
	F2:	RR	RW	RW	WW

Genotype ratio = 1:2:1 (RR:RW:WW) and

Phenotype ratio of 1:2:1 (red: pink: white)

Codominance

Co dominance occurs when the contributions of both alleles are visible in the phenotype. Co dominance where allelic products co-exist in the phenotype is different from incomplete dominance where the quantitative interaction of allele products produce an intermediate phenotype. In co dominance for example, a red homozygous flower and a white homozygous flower will produce offspring that have red and white spots. When plants of the F1 generation are self-pollinated, the phenotype and genotypic ratio of the F2 generation will be 1:2:1.

The MN blood group in humans is under the control of a pair of co-dominant alleles, L^M and L^N based based upon two genes (glycophorin A and glycophorin B) on chromosome 4.

If an individual with M blood type (L^ML^M) is married to an individual with N blood type (L^NL^N) the F1 will have MN blood type (L^ML^N) as shown below.

P1: MN blood group: L^ML^M (M)	x	L^NL^N(N)
Gametes (homogametic):	M	N
F1:	LMLN (MN)	

Multiple alleles

Multiple alleles is a type of non-Mendelian inheritance pattern that involves more than just the typical two alleles that usually code for a certain characteristic in a species. With multiple alleles, there are more than two phenotypes available depending on the dominant or recessive alleles that are available in the trait and the dominance pattern. However, only two forms of the alleles exist in a diploid individual. Why?

We will look at the most well-known ABO blood type system to help us understand how multiple alleles work. There are four principal types A, B, AB and O. There are two antigens and two antibodies that are mostly responsible for the ABO types. A and B are co-dominant and both are dominant over O. Below are the phenotypes you would expect for each genotype.

Genotype I^AI^B	=	Phenotype AB
Genotype I^AI^A	and I^AI^O =	Phenotype A
Genotype I^BI^O	= Phenotype B	
Genotype I^OI^O	= Phenotype O	

Lethal alleles

Lethal alleles are alleles that cause 100% mortality in the organism that carry them. They are usually a result of mutations in genes that are essential to

growth or development. Lethal alleles may be recessive, dominant or conditional depending on the gene or genes involved. Lethal alleles can cause death of an organism prenatally or any time after birth, though they commonly manifest early in development.

Lethal alleles were first discovered by Lucien Cuénot in 1905 while studying the inheritance of coat colour in mice. The agouti gene in mice is largely responsible for determining coat colour. The wild-type allele produces a blend of yellow and black pigmentation in each hair of the mouse. This yellow and black blend may be is referred to as 'agouti' in colour. One of the mutant alleles of the agouti gene results in mice with a much lighter, yellowish colour. When these yellow mice were crossed with homozygous wild-type mice, a 1:1 ratio of yellow and dark grey offspring were obtained. This indicated that the yellow mutation is dominant, and all the parental yellow mice were heterozygotes for the mutant allele.

Let us consider what was happening in these crosses.

e.g. Wild type genotype (agouti) = AyAy

Ayay = Yellowish colour,

if we take P1: Ayay x AyAy

Gametes: Ay/ay Ay

F1: AyAy and Ayay (in what ratio?)

selfing agouti P2: Ay ay x Ay ay (self the F1)

Gametes: Ay /ay

F2: AyAy, Ayay, Ayay, ayay

(ratio of 2:1 of agouti:black rats)

Duplicate genes

This occurs if alleles duplicated. Duplication of alleles can result from any process that result DNA replication or repair. Let us consider this by looking at a hypothetical example where by a gene controlling height is duplicated.

P1: AABB (tall) x aabb(short)
Gametes: AB ab
F1: AaBb (tall)

All the F1 generation will be tall. Now if we self F1 in a case where there are duplicate genes.

Parents 2: AaBb x AaBb
Gametes: AB/Ab/aB/ab

Using a punnet square we obtain the following

	AB	**Ab**	**aB**	**ab**
AB	AABB	AABb	AaBB	AaBb
Ab	AABb	Aabb	AaBb	Aabb
aB	AaBB	AaBb	aaBB	aaBb
ab	AaBb	Aabb	aaBb	aabb

All offspring will be tall except one. The ratio will be 15:1 of tall: short instead of the monohybrid 3:1 or dihybrid 9:3:3:1 ratio

Epistasis genes

Epistasis is the phenomenon of the effect of one gene being dependent on the presence of one or more 'modifier genes'. This is a situation where there is a gene at locus A which masks expression of another gene at locus B. Locus A suppresses the expression of the gene at locus B and it depends on whether

there is a dominant or a recessive gene on locus A. Thus, epistatic mutations have different effects in combination than individually.

I hope you will understand this easily by looking at an example of position of flowers. Let gene A be responsible for lateral flower position and B is terminal. If A and B are epistatic where A maskes the expression of B then.

Genotype AABB will give lateral flower position regardless of the presence of a dominant gene for terminal flower. If aa, a recessive gene for lateral flower is present then a terminal flower phenotype will be observed. Terminal flower will only be expressed where aa is recessive. If loci A is heterozygous, that is there is Aa at this locus, lateral flower position will be the phenotype.

Complementary genes

This is a situation in which two independent pairs of non-allelic genes neither of which is functional without the other. These are genes that produce an observable effect in an organism only in conjunction with another gene. If seed texture is controlled by two alleles A and B and that they two alleles are complementary to each other it means if AABB is the multilocus genotype the seed will be smooth. The following genotypes will give wrinkled seed because the other locus is recessive and hence does not complete the other allele.

i.e.	AA	+	BB	= smooth seed
	AA	+	bb	= wrinkled seed
	aa	+	BB	= wrinkled seed
	aa	+	bb	= wrinkled seed

Polygenic inheritance

These are multiple genes determining the degree of a characteristic. One of a group of genes in which the number of the genes present collectively determines the extent of a characteristic. The term “polygenic inheritance” is

used to refer to the inheritance of quantitative traits, traits which are influenced by multiple genes. In addition to involving multiple genes, polygenic inheritance is also influenced by the environment in someone's development.

Because many traits are spread out across a continuum, rather than being divided into black and white differences, polygenic inheritance helps to explain the way in which these traits are inherited and focused. A related concept is pleiotropy, an instance where one gene influences multiple traits.

A typical example would be human height which is controlled by a number of gene and depending on the number of genes contributing to the character. The individual has a range of possible heights not just discrete values.

Linkage

Genetic linkage is the tendency of alleles that are close together on a chromosome to be inherited together during the meiosis phase of sexual reproduction. Genes whose loci are nearer to each other are less likely to be separated onto different chromatids during chromosomal crossover, and are therefore said to be genetically linked. In other words, the nearer two genes are on a chromosome, the lower is the chance of a swap occurring between them, and the more likely they are to be inherited together.

Genes that are on the same chromosome are called linked genes. These genes are associated in their inheritance, they are inherited together and in the same position. The linkages are found either in sex cells/sex chromosomes or autosomal cells/autosome chromosomes.

Sex linkage

Morgan observed that inheritance of a characteristic depends on whether the characteristic is on a male or a female organism. He made these observations when he was studying inheritance pattern of eye colour in fruit flies.

When he crossed wildtype female fruit flies and red eyed males he observed that all females offspring were red eyed but the males had wild type eyes. The pattern was different when he crossed red eyed females and wild type eyed males. In this crossed he observed that all the offspring had red eyes.

These crosses can be represented as shown below.

Parents:	Female (wildtype)	x	Male (red eyes)
	X^cX^c		X^CY
Gametes:	X^C		X^C, Y
F1:	$X^C X^c$, X^cY, X^C, X^C, X^cY		

(all females had red eyes while males had wild type)

When the characteristics are reversed the following phenotypes are observed.

P1:	Female (red eyes)	x	Male (wild type)
	$X^C X^C$		X^cY
F1:	$X^C X^C$, X^CX^C, X^CY, X^CY		
	(all of them had red eyes)		

There are several examples of sex linked characteristics in human beings however for this purpose it should suffice to mention the following characteristic which most of you are familiar with.

1) colour blindiness associated with X
2) haemophilia, mainly suffered by females
3) baldnes

Autosomal linkage

On the other hand there are two types of autosomal linkages; complete and partial autosomal linkages. Genes that are linked completely do not assort themselves independently as a result they are inherited in a block.

Partial linkages come as a result of crossing over at meiosis to make new combinations with other genes called recombinants. The frequency of recombinants is used to estimate the distance between genes.

You will recall that we noted that genes that are far apart have a higher probability of crossing over than genes that are close together. Morgan proposed a standard percentage combinations that 1% of recombinants represent 1 map unit. This you can understand easily when you consider that when you have a 1% frequency of recombinants then the two genes are one map unit apart. The map unit was named after him as centimorgan hence 1% means the two genes are 1 cM (centimorgan) apart.

Let us look at an example from a typical cross of coloured smooth seeds and coloured wrickled seeds from which the following combinations were found

Coloured smooth seeds 48.25%
Coloured wrickled seeds 48.25%
Colourless wrickled seeds 1.75%
Colourless smooth seeds 1.75%

Total percentage of recombinants for this experiment is 1.75% + 1.75% = 3.5%. We can therefore conclude from this data that the gene for colour and texture are 3.5cM apart

Mutations

We will start by defining mutations as sudden changes to chromosomal DNA that results into some heritable characteristics. Factors that cause mutations also known as mutagens or mutagenic factors have potential of denaturing proteins and DNA. Mutagenic factors can be categorised as radiation which includes X rays, Alpha rays, gamma rays and beta rays. Ultra violet (UV) light is also a radiation that is a mutagen. Another category of mutagenic factors are oxidation. Oxidation factors are Hydrogen peroxide H_2O_2 and Mustard gas.

There are two types of mutation – gene and chromosome mutation affecting the gene and the chromosome respectively

Types of gene mutations

1) Substitution- nucleotide is replaced by another. The nucleotides work in triples and they code for a particular amino acid. When one nucleotide is changed the amino acid changes and it affects its function.
2) Insertion/addition- nucleotide(s) are added to a DNA strand
3) Inversion- nucleotides are cut off and pasted to the same strand in an inverted way
4) Deletion- the nucleotide(s) is cut and disappear

These mutations have been illustrated in figure 3.6 below.

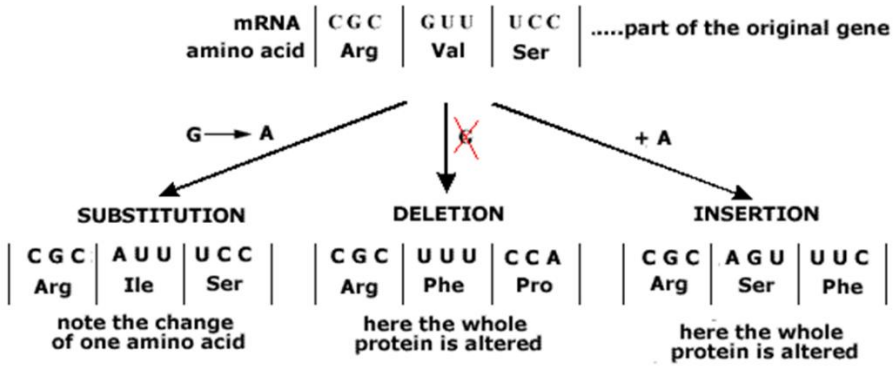


Figure 3.6:. Types of gene mutations

Types of chromosome mutations

1) Substitution- a small part of the chromosome is cut and lost but is replaced by another part
2) Translocation/ addition- this involves transfer of a chromosome part to another chromosome
3) Inversion- part of a chromosome is cut and joins the same chromosome after flipping over.
4) Deletion- when part of a chromosome disappears
5) Duplication – a section of a chromosome is repeated
6) Mutations involving change in chromosome number- non- disjunction at anaphase of mitosis or meiosis, this may result in one cell having less or more than 2n

   e.g. Male XY female XX . Non disjuction in male may result in XO = tuner female, XXY = super male or XXX = super female

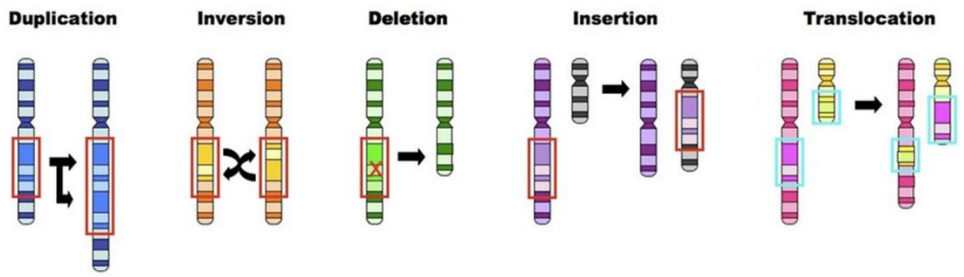


Figure 3.7: types of chromosomal mutations

3.4 Population genetics

Until now we have focused our attention on genes within individuals and their related family members. The field of population genetics is concerned with understanding changes in genetic variations within a group of individuals over time. Population genetics seeks to know the extent of genetic variations within populations.

Different alleles within a population can be identified by phenotypic effects (for example colour, size, shape), variations in molecular structure (such as protein or DNA).

Population genetics is an extension of our understanding of Mendel's laws of inheritance and molecular genetics. Conceptually, all of the alleles of every gene in a population make up the gene pool. Population genetics study the genetic variation within the gene pool and how such variation changes from one generation to the next.

We will define a population as a group of individuals of the same species that occupy the same region and can be interbreed with one another. A large population usually is composed of smaller groups called local populations or demes.

Population genetics is concerned with allele and genotype frequencies. In population genetics we want to understand the prevalence of polymorphic genes within populations. The goal is to identify the causative factors that govern changes in genetic variation. Allele frequencies can be used to make inferences about mating patterns, selection, migration etc.

Two fundamental calculations are central to population genetics:-

$$Allele\ frequency = \frac{Number\ of\ copies\ of\ an\ allele\ in\ a\ population}{Total\ number\ of\ alleles\ for\ that\ gene\ in\ a\ population}$$

$$Genotype\ frequency = \frac{Number\ of\ individuals\ with\ a\ particular\ genotype\ in\ a\ population}{Total\ number\ of\ individualss\ in\ a\ population}$$

Hardy and Weinberg proposed that allele frequencies in a population remain constant—that is, they are in equilibrium—from generation to generation unless specific disturbing influences are introduced. These disturbing factors include non-random mating, mutations, selection, limited population size and migration.

The Hardy-Weinberg equation is a mathematical equation that can be used to calculate the genetic variation of a population at equilibrium. To explore the Hardy-Weinberg equation, we can examine a simple genetic locus at which there are two alleles, A and a. The Hardy-Weinberg equation is expressed as:

$$p2 + 2pq + q2 = 1$$

where p is the frequency of the "A" allele and
q is the frequency of the "a" allele in the population.

The formula is sometimes written as (p2) + (2pq) + (q2)= 1, representing the trivial fact that frequencies must add up to one. The final three possible genotypic frequencies in the offspring become: f(AA) = p2, f(Aa) =pq and f(aa) = q2. These frequencies are called Hardy–Weinberg frequencies (or Hardy–Weinberg proportions).

In the equation, p2 represents the frequency of the homozygous genotype AA, q2 represents the frequency of the homozygous genotype aa, and 2pq represents the frequency of the heterozygous genotype Aa. In the simplest case of a single locus with two alleles: the dominant allele is denoted A and the recessive a and their frequencies are denoted by p and q; freq(A) = p; freq(a) = q; p + q = 1.

If the p and q allele frequencies are known, then the frequencies of the three genotypes may be calculated using the Hardy-Weinberg equation. In

population genetics studies, the Hardy-Weinberg equation can be used to measure whether the observed genotype frequencies in a population differ from the frequencies predicted by the equation. If the population is in equilibrium, then we will have freq(AA) = p2 for the AA homozygotes in the population, freq(aa) = q2 for the aa homozygotes, and freq(Aa) = 2pq for the heterozygotes.

Let us consider data that was collected in Britain on MN blood system. The study found out that the frequency of M to be 0.54 while the frequency of N was 0.46. From this data we can calculate the frequency of MM, MN and NN blood types.

You will notice that the frequency of M is given as p = 0.54 and the frequency of N is given as q = 0.46. Assuming that the population is in Hardy-Weinberg equilibrium then expected genotypic frequencies will therefore be

MM	**MN**	**NN**
p2	2pq	q2
0.294	0.496	0.209

Now let us assume that 1000 individuals were sampled in the following generation and there were 298 individuals with genotype M, 489 with MN genotype and 213 were NN. We can measure if this population is in Hardy Weinberg equilibrium using Chi square test.

We start by calculating the expected genotypes under Hardy Weinberg equilibrium. Then we can proceed to test if the data is in agreement with Hardy Weinberg equilibrium.

Genotypes	MM	MN	NN
Expected	294	496	209
Observed	298	489	213

The table below summaries the steps that we follow to test whether the frequencies are in agreement with Hardy-Weinberg equilibrium. We start by subtracting the expected from the observed and to remove the negatives the value is squared. Then in the next column you divide this value by the expected.

In the column the values are added to give a χ^2 number which you look up in the χ^2 table. Refer to the practical module for more practise and examples.

Observed (O)	Expected (E)	$(O\text{-}E)^2$	$\frac{(O-E)^2}{E}$
298	294	16	0.054
489	496	49	0.099
213	209	16	0.077
			$\chi^2 = 0.230$

3.5 Principles of genetic evolution

We will define evolution as a change in heritable characteristics of biological populations over successive generations. Evolutionary processes give rise to biodiversity at every level of biological organisation.

It is believed that all life on Earth shares a common ancestor known as the last universal common ancestor (LUCA) which lived approximately 3.5–3.8 billion years ago. Repeated formation of new species (speciation), change within

species (anagenesis), and loss of species (extinction) throughout the evolutionary history of life on Earth are demonstrated by shared sets of morphological and biochemical traits. These shared traits are more similar among species that share a more recent common ancestor, and can be used to reconstruct a biological "tree of life" based on evolutionary relationships (phylogenetics) using both existing species and fossils.

In the mid-19th century, Charles Darwin formulated the scientific theory of evolution by natural selection, published in his book On the Origin of Species (1859). Evolution by natural selection is a process demonstrated by the observation that more offspring are produced than can possibly survive, along with three facts about populations: phenotypic variation, differential fitness and heritability of fitness. Thus, in successive generations members of a population are replaced by progeny of parents better adapted to survive and reproduce in the biophysical environment in which natural selection takes place. Natural selection, including sexual selection, is the only known cause of adaptation but not the only known cause of evolution. Other, non-adaptive evolutionary processes include mutation, genetic drift and gene migration.

Over a long period of time, precise mechanisms of reproductive heritability and origin of new traits have remained a mystery. However, in the 1920s and 1930s a modern evolutionary synthesis connected natural selection, mutation theory and Mendelian inheritance. Modern evolutionary synthesis explains how evolution acts on patterns of change within individuals. Some of the mechanisms include heredity, variations, mutation, recombination, gene flow, natural selection, genetic drift, genetic hitchhiking, adaptation, coevolution, cooperation, speciation, extinction.

Evidence of genetic evolution

Scientist have demonstrated that organisms have evolved from a common ancestor. This evidence constructs the theoretical framework on which evolutionary theory rests, demonstrates that evolution does occur and is able to show the natural processes that led to the emergency of earth's biodiversity. This evidence supports the modern evolutionary synthesis.

1. Comparative physiology and biology where strongest evidence comes from comparative sequence analysis which examines relationship between DNA or protein sequences of different species and used in generating phylogenetic trees.
2. Comparative anatomy in which groups of organism reveal basic similarities of certain structural features. This has been demonstrated by atavism which is the reappearance of lost characters typical of ancestors and seen in recent ancestors, similarity in embryogenesis, homologous structures and divergent adaptive evolution, nested hierarchies in classification, vestigial structures
3. Paleontology which is the evidence obtained from the study of past life based on fossil records and their relation to different geologic time periods. However, because of limitations inherent in the record, there are not fine scales of intermediate forms between related groups of species.
4. Biogeography in which there is evidence that organisms are adapted to their environment. Data about the presence or absence of species on various continents and islands (biogeography) can provide evidence of common descent and shed light on patterns of speciation. Further evidence can be found in continental distribution, island biogeography, endemism, adaptive radiations and rings species.

5. Artificial selection demonstrates the diversity that can exist among organisms that share a relatively recent common ancestor. Experimental evolution uses controlled experiments to test hypotheses and theories of evolution.
6. Speciation is the evolutionary process by which new biological species arise.
7. Computer science allows mathematical modelling that iteration of self-changing complex systems to be studied, allowing a mathematical understanding of the nature of the processes behind evolution; providing evidence for the hidden causes of known evolutionary events. The evolution of specific cellular mechanisms like spliceosomes that can turn the cell's genome into a vast workshop of billions of interchangeable parts that can create tools that create tools that create tools that create us can be studied for the first time in an exact way.

Unit 1 activity

Unit 1 Summary

Unit test

1. The ability to taste a bitter chemical, phenylthiocarbamide (PTC), is due to a dominant gene. Use T and t to symbolize the two alleles of this gene. a. What is the genotype of a non-taster? What are the possible genotypes of a taster? b. Could a person with two tasters as parents be a non-taster? How?

2. In the fruit fly Drosophila melanogaster, vestigial wings and hairy body are produced by two recessive genes located on different chromosomes. The

normal alleles, long wings and hairless body, are dominant. Give the genotype and phenotype of F1 progeny obtained from a cross between a vestigial-winged, hairy male and a normal, homozygous female. If the F1 from this cross are permitted to mate randomly among themselves, what phenotypic ratio would be expected in the F2 generation?

3. In humans, sickle-cell anemia is caused by a recessive lethal allele Hbs; individuals who are HbaHbs have sickle-cell trait, but are healthy. a. What is the probability of two heterozygous individuals giving birth to a child with sickle-cell trait? b. What is the probability of two heterozygous individuals giving birth to a child with sickle-cell anemia? c. If a normal HbaHba individual receives a blood transfusion from a HbaHbs individual (heterozygous for sickle cell) what are the chances that the HbaHba man and his HbaHba wife will have HbaHbs children?

4. A study on blood types in a population found the following genotypic distribution among the people sampled: 1101 were MM, 1496 were MN and 503 were NN. Calculate the allele frequencies of M and N, the expected numbers of the three genotypic classes (assuming random mating). Using X2, determine whether or not this population is in Hardy-Weinberg equilibrium.

5. The compound phenylthiocarbamide (PTC) tastes very bitter to most persons. The inability to taste PTC is controlled by a single recessive gene. In the American white population, about 70% can taste PTC while 30% cannot (are non-tasters). Estimate the frequencies of the Taster (T) and nontaster (t) alleles in this population as well as the frequencies of the diploid genotypes.

Unit answers

Reading assignment/List